
CMPE 297-01 Computer Vision

Light and Shading

Fall 2026, DMH 348



P. Claesz, [Still Life with a Skull and a Writing Quill](#), 1628

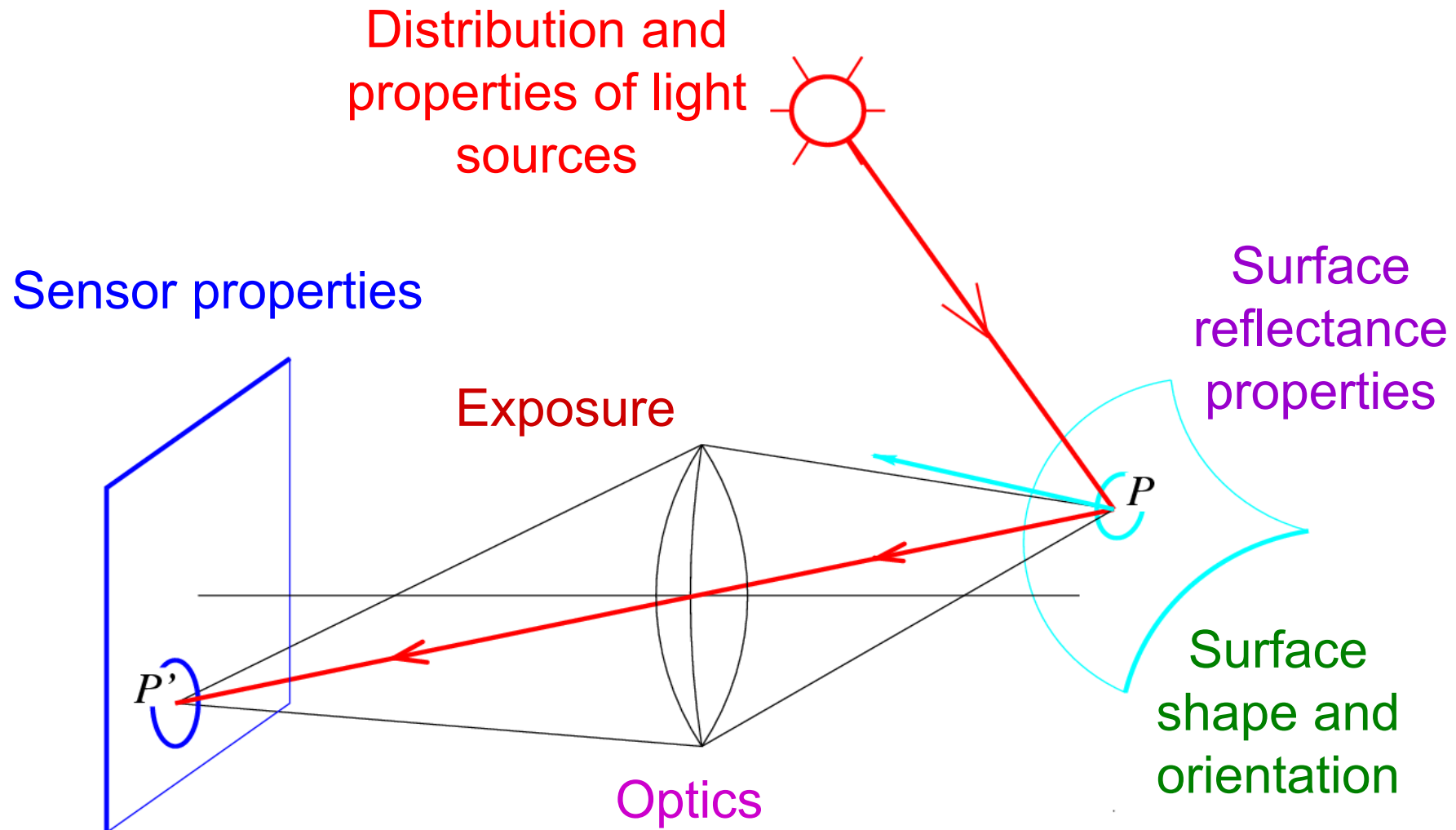
Lecture Outline

- Small taste of radiometry
- In-camera transformation of light
- Reflectance properties of surfaces
- Diffuse and specular reflection
- Shape from shading
- Estimating direction of light sources

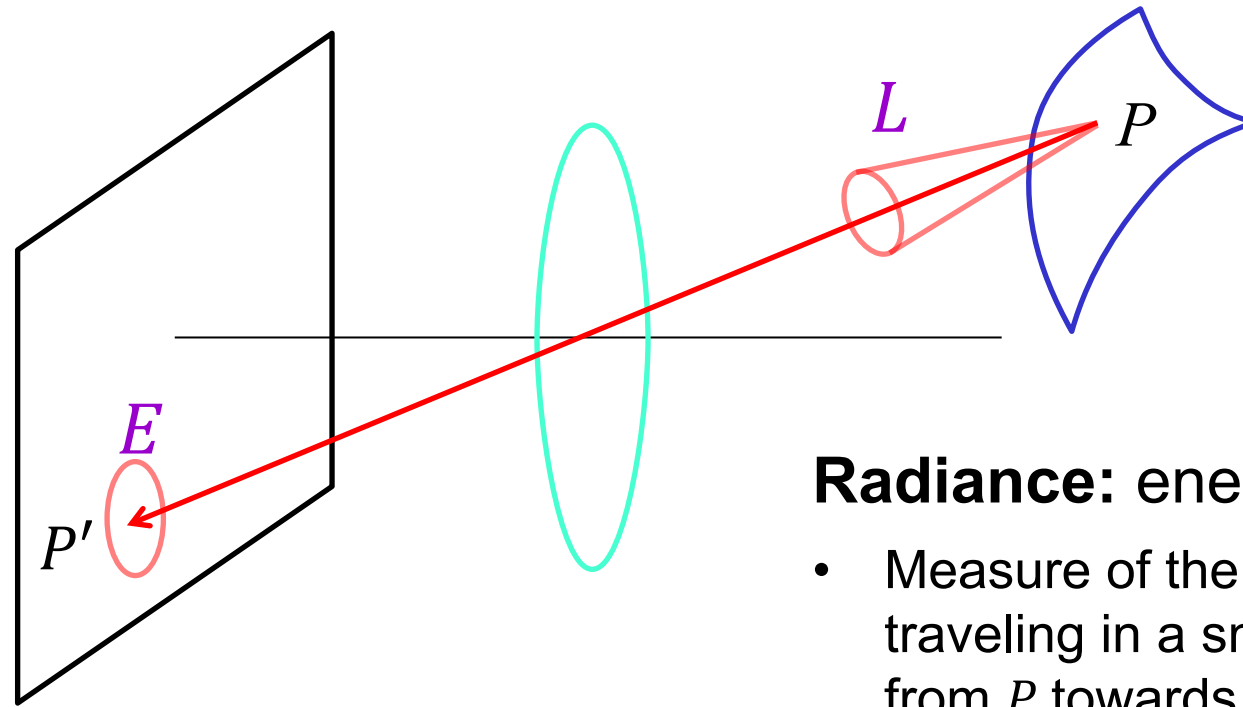
Small Taste of Radiometry

Image Formation

- What determines the *brightness* of an image pixel?



Radiometry of Image Formation



Irradiance: energy arriving at a surface

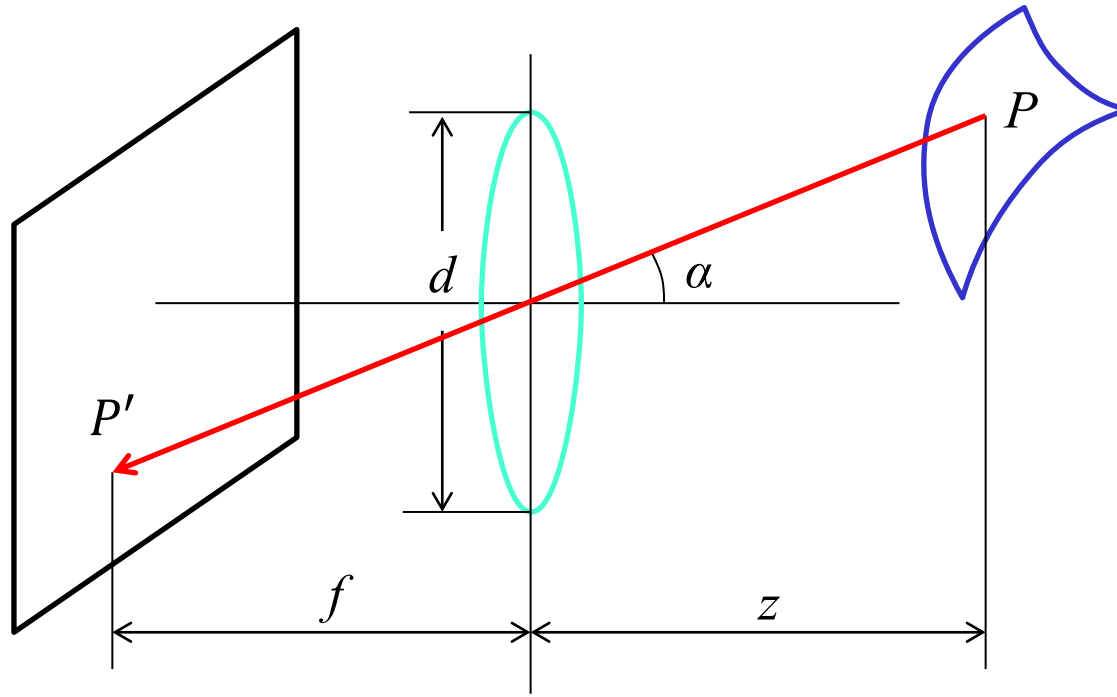
- Incident power per unit area (not foreshortened)
- Units: Watts per square meter

Radiance: energy carried by a ray

- Measure of the density of photons traveling in a small cone of directions from P towards P'
- Power per unit area perpendicular to the direction of travel, per unit solid angle
- Units: Watts per square meter per steradian

What is the relationship between E and L ?

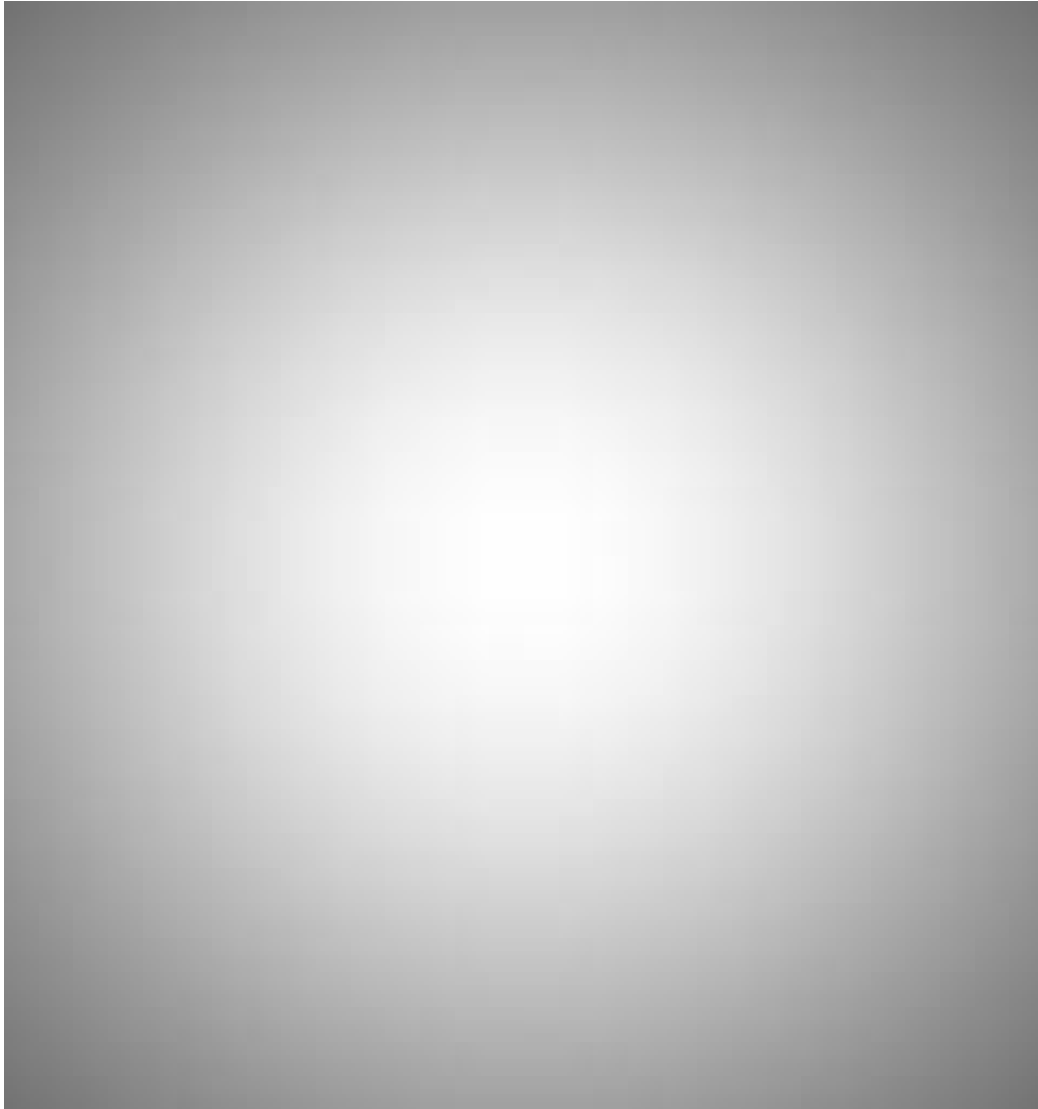
Fundamental Radiometric Relation



$$E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L$$

- Image irradiance (E) is linearly related to scene radiance (L)
- Irradiance is *directly* proportional to the area of the lens ($\frac{\pi d^2}{4}$) and *inversely* proportional to the squared distance between the lens and the image plane (f)
- The irradiance decreases as the angle between the viewing ray and the optical axis (α) increases

Fundamental Radiometric Relation

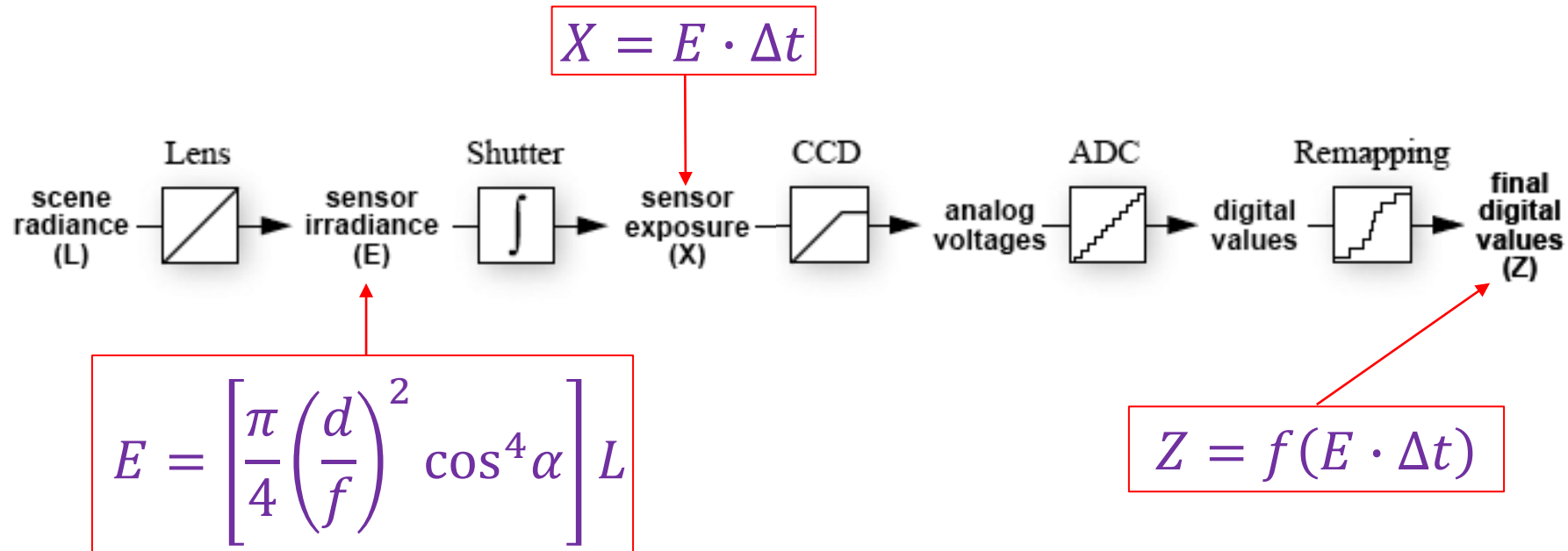


$$E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L$$

S. B. Kang and R. Weiss. [Can we calibrate a camera using an image of a flat, textureless Lambertian surface?](#)
ECCV 2000

In-Camera Transformation of Light

From Light Rays to Pixel Values

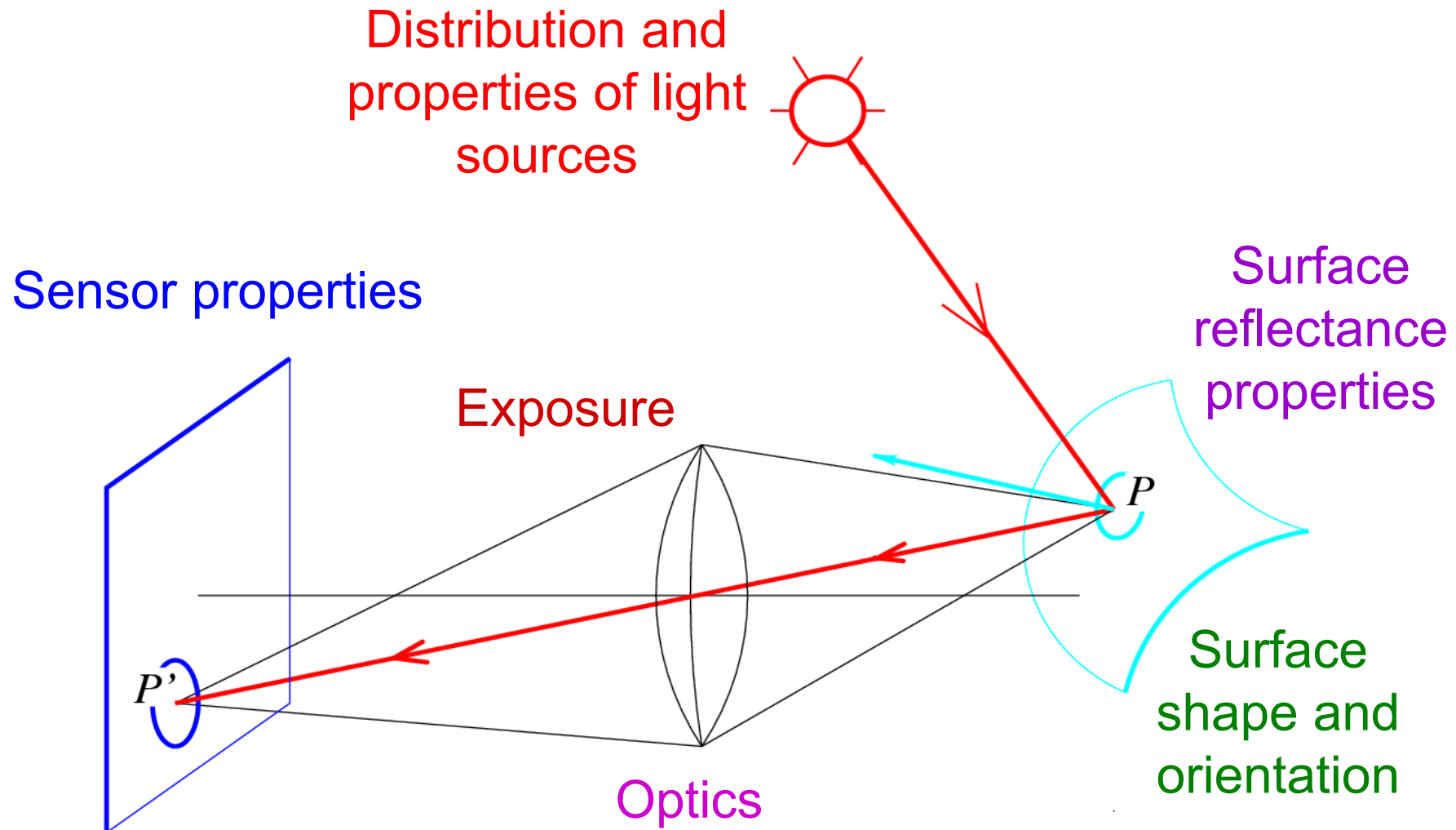


- **Camera response function:** the mapping f from irradiance to pixel values
 - Needed for applications like estimation of scene reflectance properties, creating high dynamic range (HDR) images
 - For further reading: M. Brown, [Understanding the In-Camera Image Processing Pipeline for Computer Vision](#), CVPR 2016 Tutorial

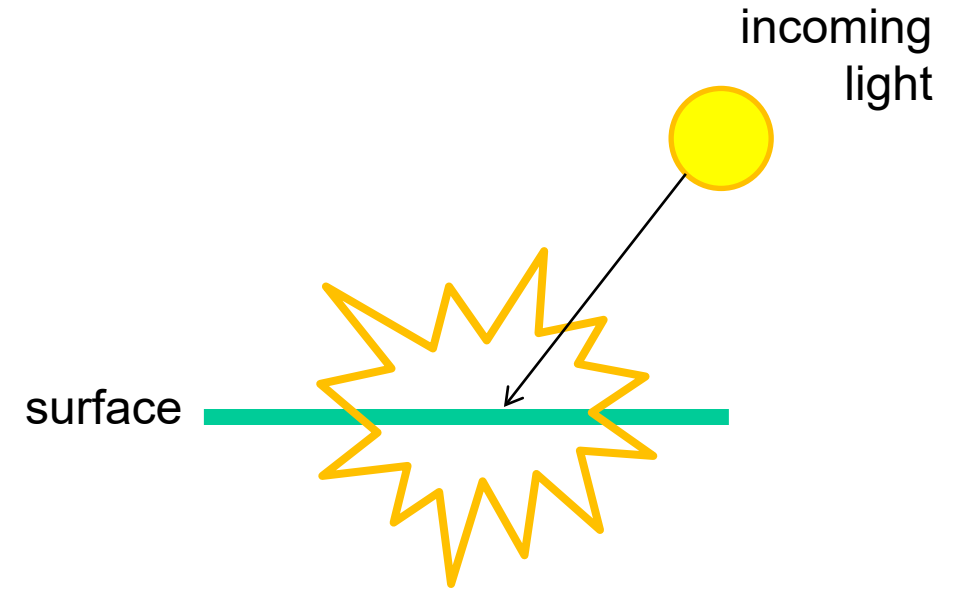
Reflectance Properties of Surfaces

Recall: Image Formation

- What determines the brightness of an image pixel?

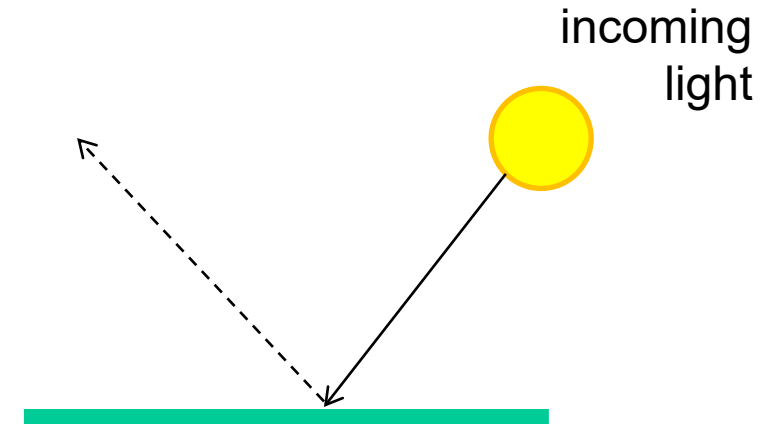


What can happen to light when it hits a surface?

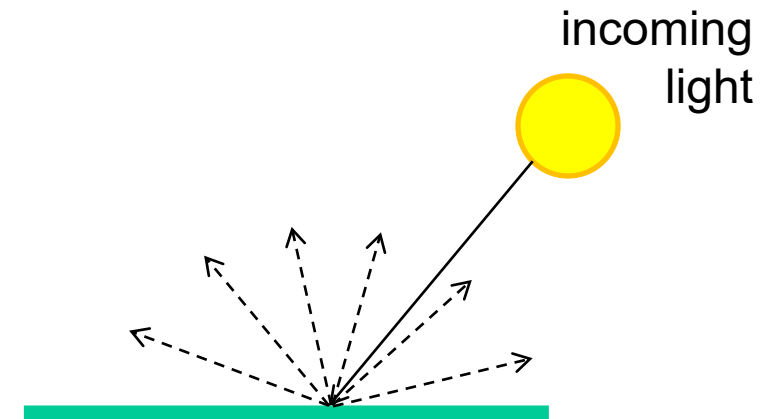


Basic Models of Reflection

- **Specular reflection:** light is reflected about the surface normal



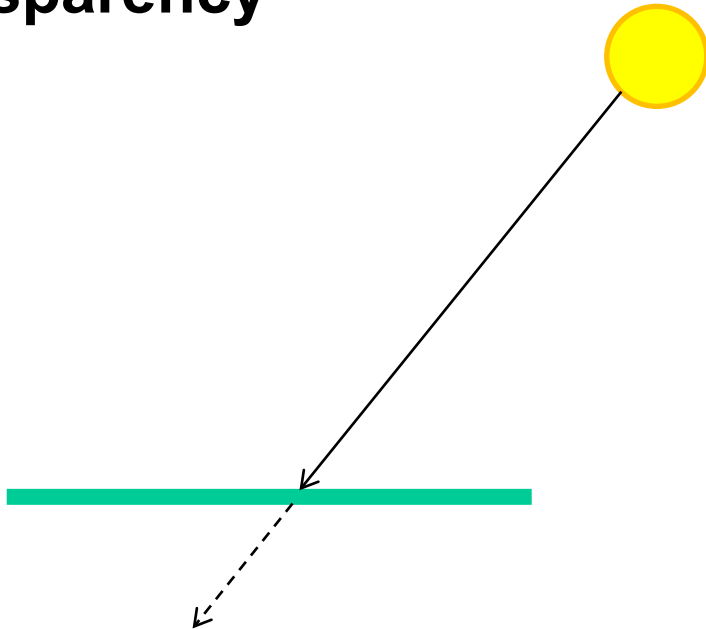
- **Diffuse reflection:** light scatters equally in all directions



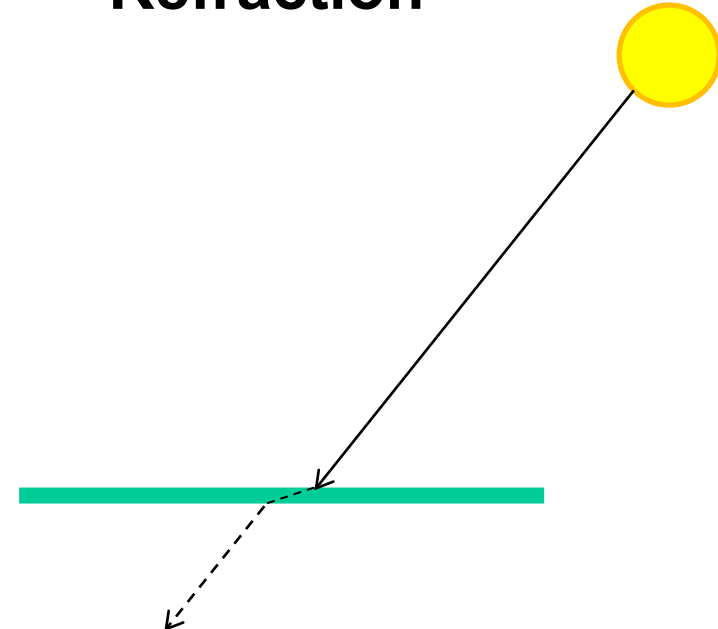
Other Possible Effects



- **Transparency**

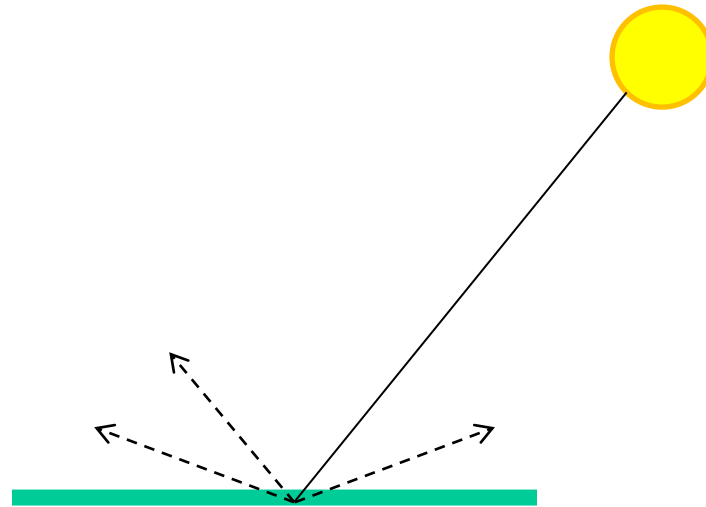
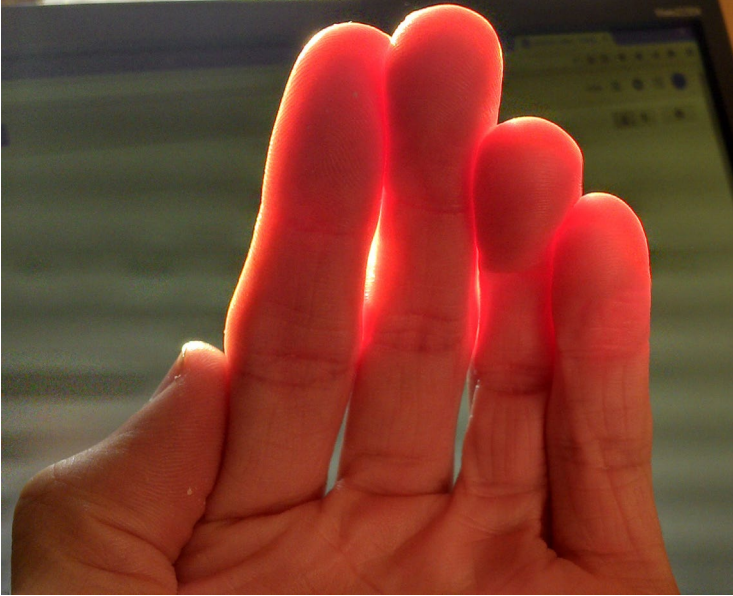


- **Refraction**



Other Possible Effects

- Subsurface scattering

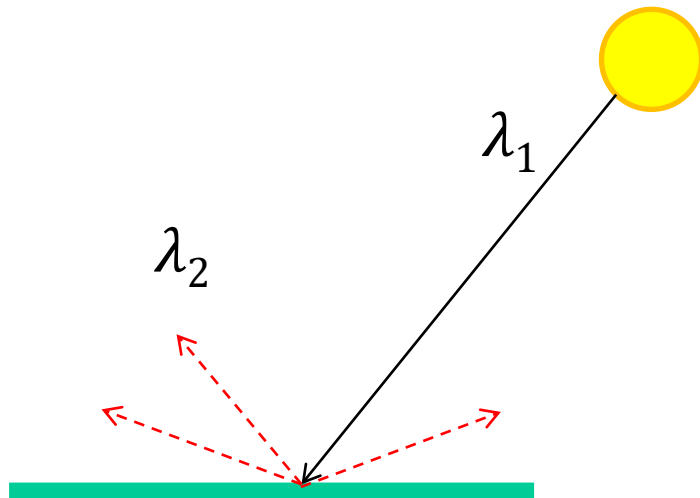


Slide from D. Hoiem

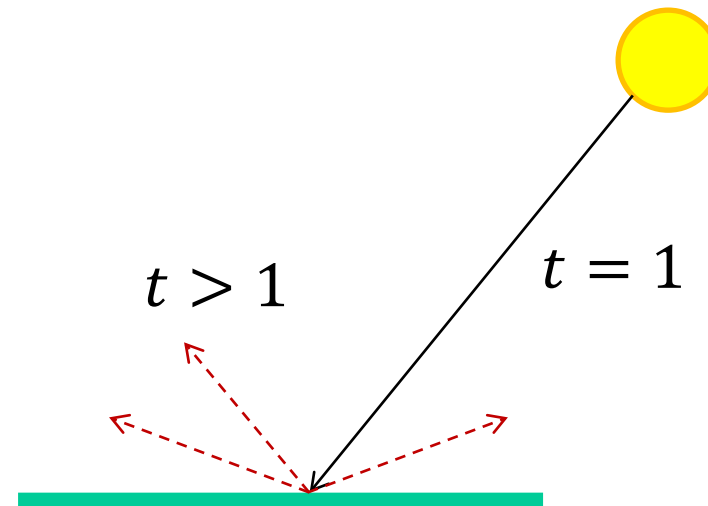
[Image source](#)

Other Possible Effects

- Fluorescence



- Phosphorescence

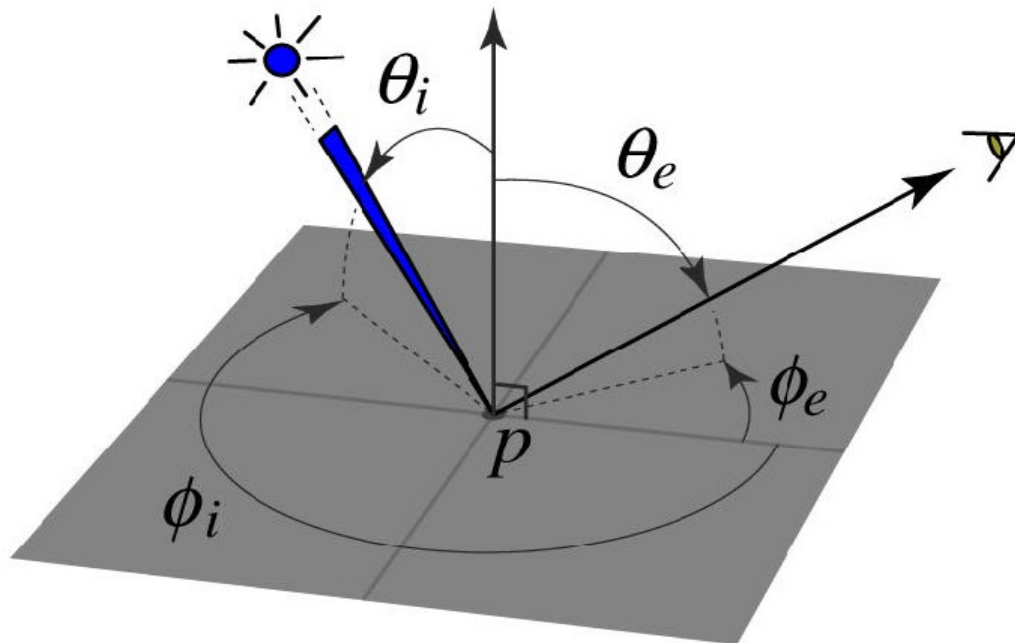


Slide from D. Hoiem

[Image source](#)

Bidirectional Reflectance Distribution Function (BRDF)

- How bright a surface appears when viewed from one direction when light falls on it from another
- Definition: ratio of the radiance in the emitted direction to irradiance in the incident direction



Function of (at least) four parameters: incident and outgoing θ, ϕ

Bidirectional Reflectance Distribution Function (BRDF)

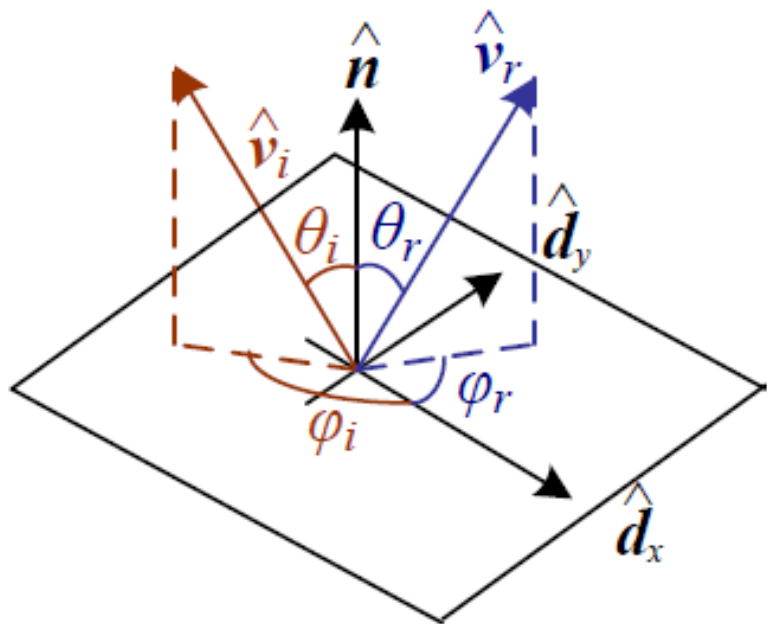
- How bright a surface appears when viewed from one direction when light falls on it from another
- Definition: ratio of the radiance in the emitted direction to irradiance in the incident direction
- Can be incredibly complicated!



Amount of Light Exiting a Surface

$$L_r(\hat{\mathbf{v}}_r; \lambda) = \int L_i(\hat{\mathbf{v}}_i; \lambda) f_r(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}}; \lambda) \cos^+ \theta_i d\hat{\mathbf{v}}_i,$$

$$\cos^+ \theta_i = \max(0, \cos \theta_i).$$



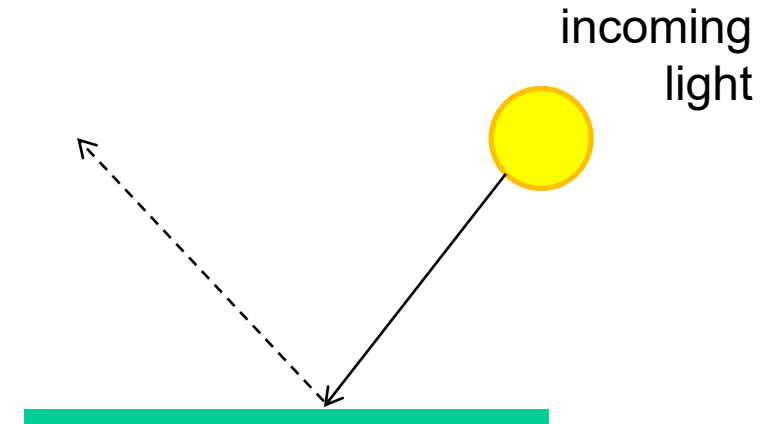
If the light sources are discrete (a finite number of point light sources),

$$L_r(\hat{\mathbf{v}}_r; \lambda) = \sum_i L_i(\lambda) f_r(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}}; \lambda) \cos^+ \theta_i.$$

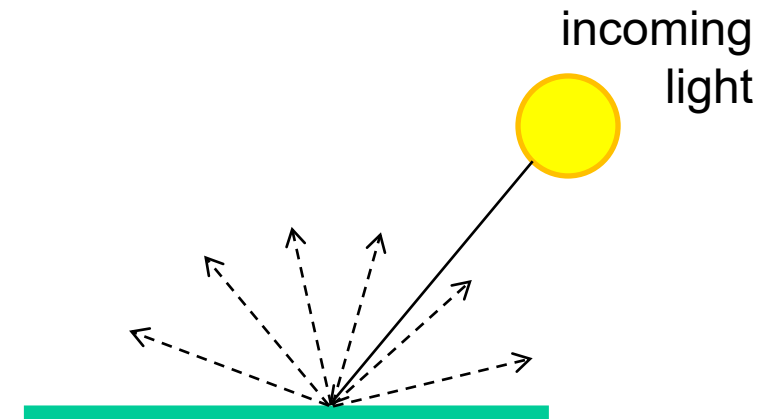
Diffuse and Specular Reflection

Basic Models of Reflection in Detail

- **Specular reflection:** light is reflected about the surface normal

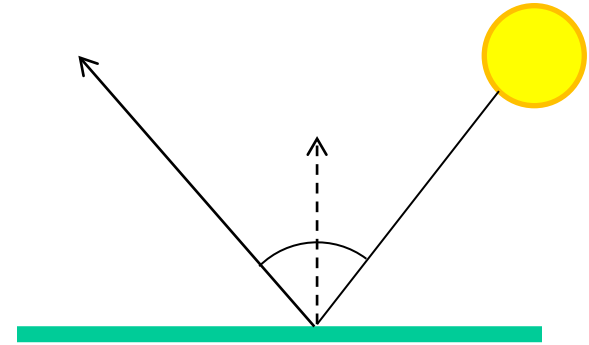


- **Diffuse reflection:** light scatters equally in all directions



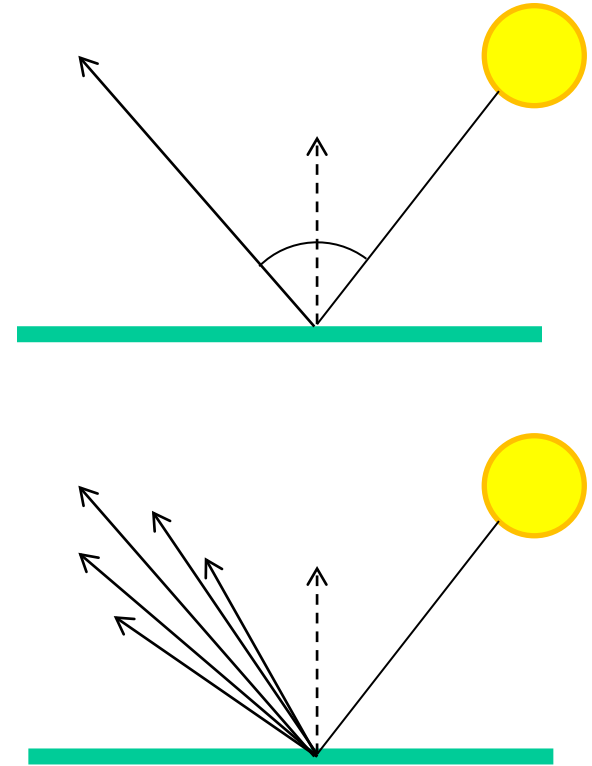
Specular Reflection

- Radiation arriving along a source direction leaves along the **specular direction** (source direction reflected about normal)



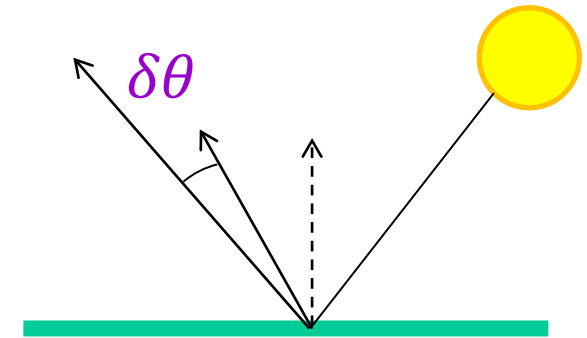
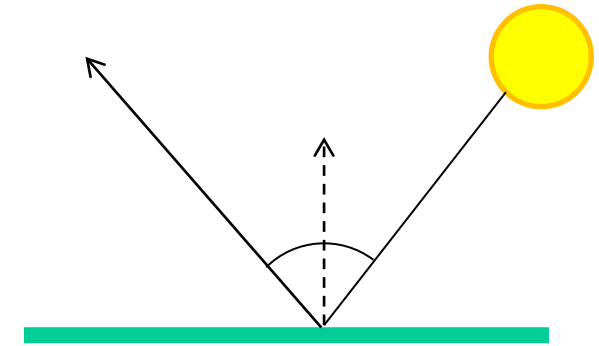
Specular Reflection

- Radiation arriving along a source direction leaves along the **specular direction** (source direction reflected about normal)
- On real surfaces, energy usually goes into a “lobe” of directions

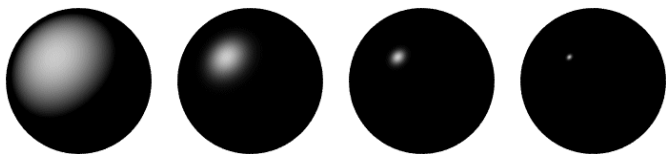


Specular Reflection

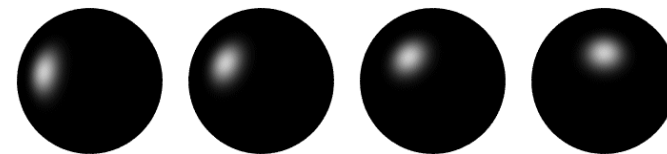
- Radiation arriving along a source direction leaves along the **specular direction** (source direction reflected about normal)
- On real surfaces, energy usually goes into a “lobe” of directions
- **Phong model:** reflected energy falls off with $\cos^n(\delta\theta)$



Changing the exponent



Moving the light source



Specular Reflection

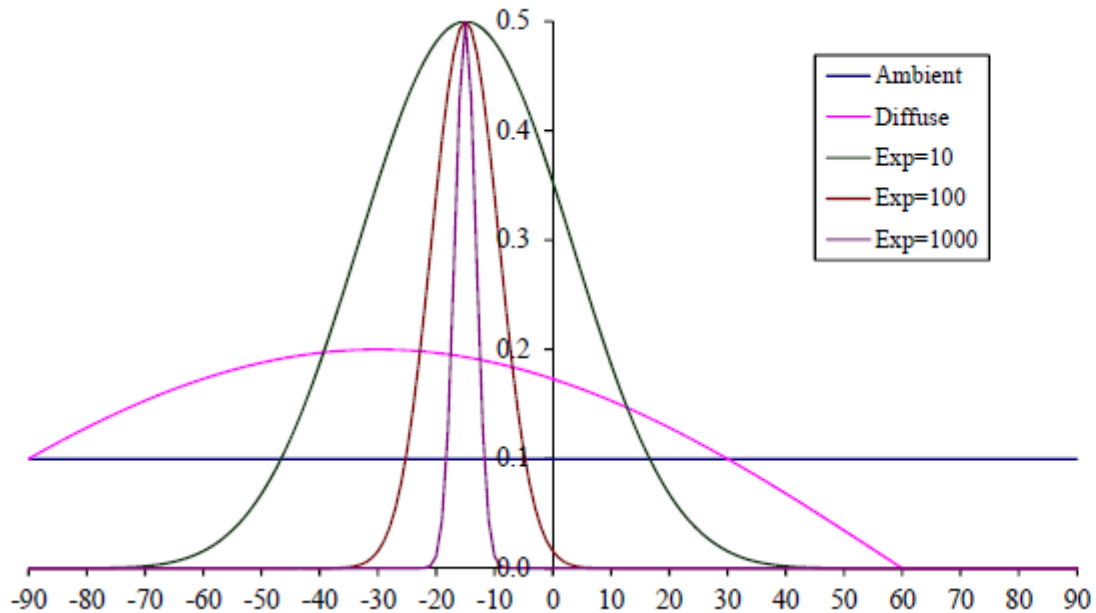
Phong Model

$$f_s(\theta_s; \lambda) = k_s(\lambda) \cos^{k_e} \theta_s,$$

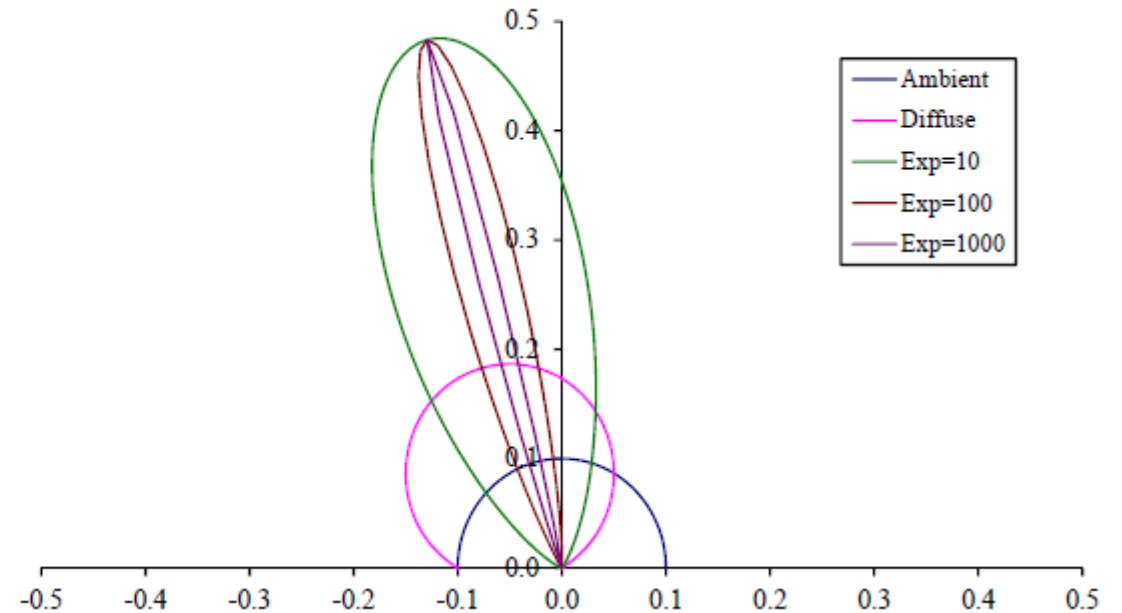
Torrance and Sparrow Model

$$f_s(\theta_s; \lambda) = k_s(\lambda) \exp(-c_s^2 \theta_s^2).$$

Cross-Section Through a Phong Shading Model BRDF



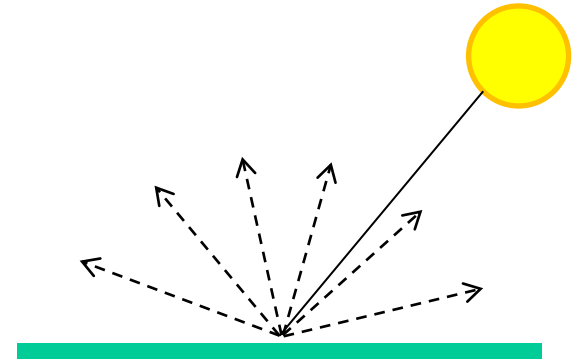
Component values as a function of angle away from the surface normal



Polar plot

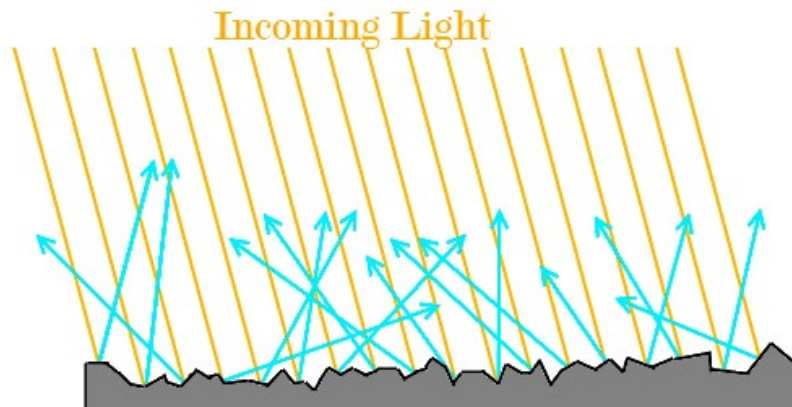
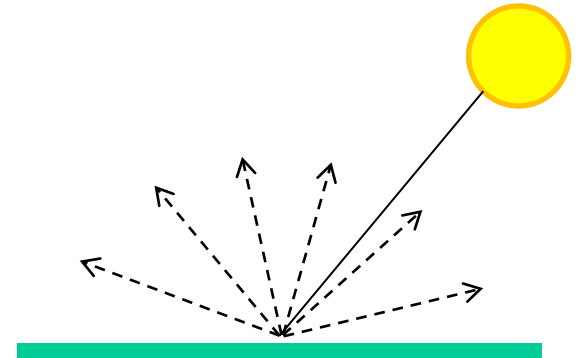
Diffuse Reflectance

- Light scatters equally in all directions
 - E.g., brick, matte plastic, rough wood



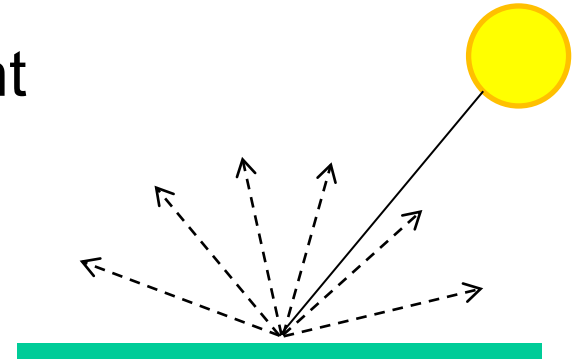
Diffuse Reflectance

- Light scatters equally in all directions
 - E.g., brick, matte plastic, rough wood
- This happens because of *microfacets* that scatter incoming light randomly

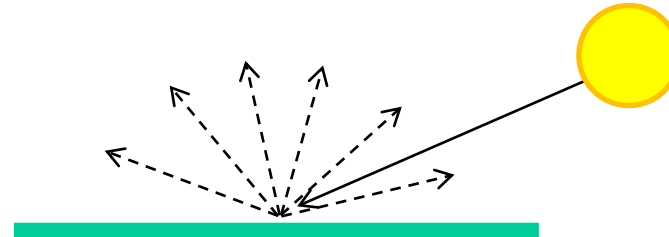
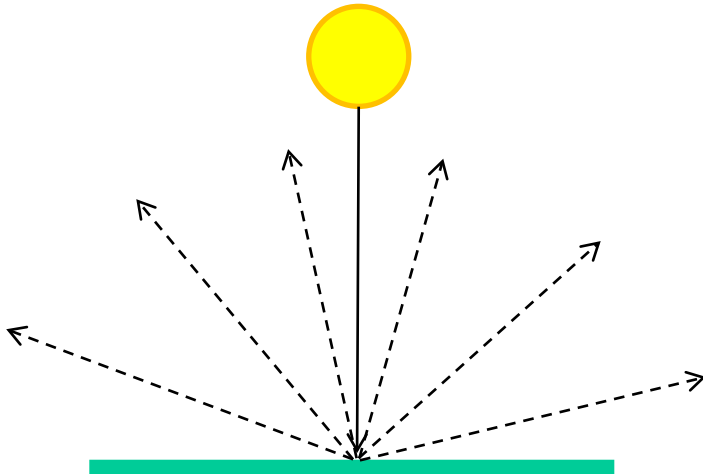


Diffuse Reflectance

- Light scatters equally in all directions
 - For a fixed incidence angle, BRDF is constant



- What if we change the incidence angle?



Why do we care about diffuse reflectance?

Same lighting, as close as possible camera settings, but different **camera position**



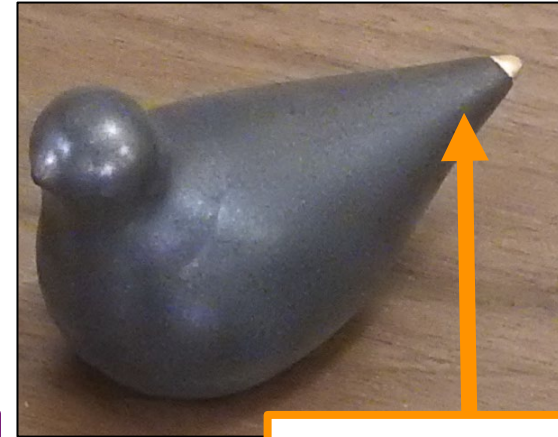
Diffuse



Same appearance



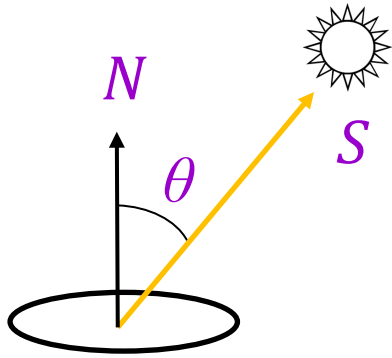
Specular



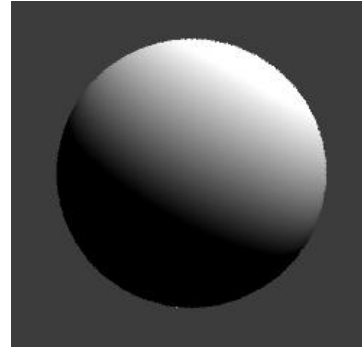
Totally different appearance



Diffuse Reflectance: Lambert's law



$$\begin{aligned} I &= \rho (S \cdot N) \\ &= \rho \|S\| \cos \theta \end{aligned}$$



I : reflected intensity (technically: *radiosity*, or total power leaving the surface per unit area)

ρ : albedo (fraction of incident irradiance reflected by the surface)

S : direction of light source (magnitude proportional to intensity of the source)

N : unit surface normal

Shape from Shading

Photometric Stereo, or Shape from Shading

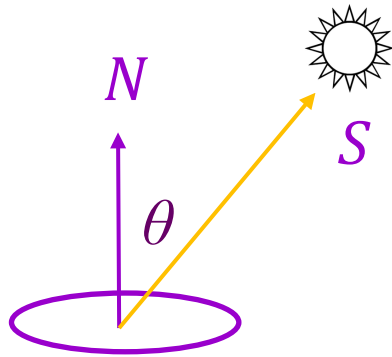
- Can we reconstruct the shape of an object based on shading cues?



Luca della Robbia,
Cantoria, 1438

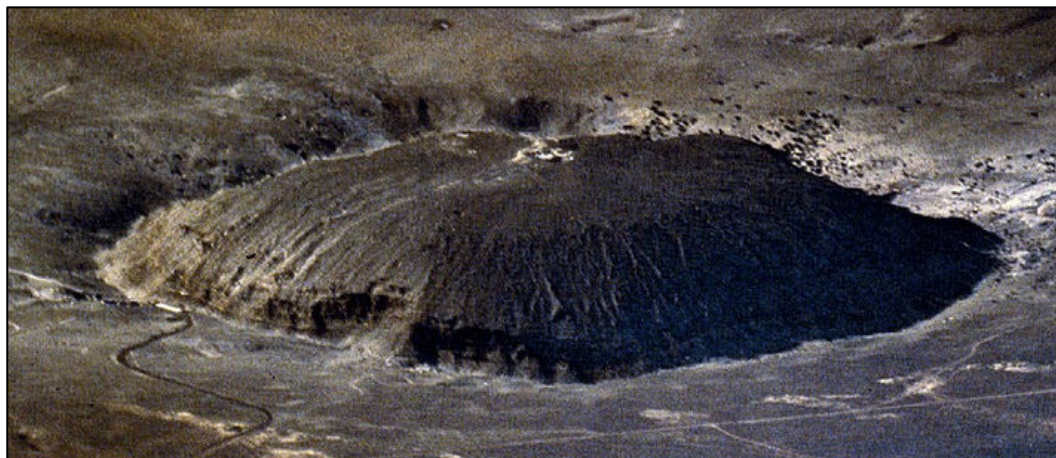
Photometric Stereo, or Shape from Shading

- Can we reconstruct the shape of an object based on shading cues?
- Assuming a Lambertian object, given the image intensity (I), can we recover the light source direction (S) and the surface normal (N)?
- Can we do this from a single image?



$$\begin{aligned} I &= \rho (S \cdot N) \\ &= \rho \|S\| \cos \theta \end{aligned}$$

Shape from Shading Ambiguity



Shape from Shading Ambiguity

- Humans assume light from above (and the blueness also tells you distance)

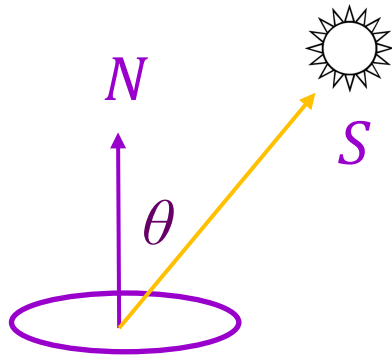
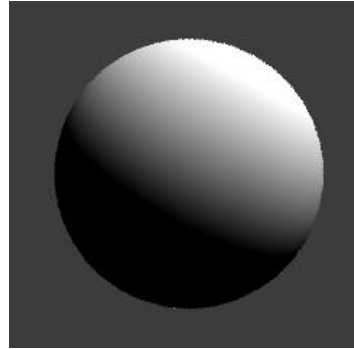


Last time: Light and Shading

- Small taste of radiometry
- In-camera transformation of light
- Reflectance properties of surfaces
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Review: Lambert's Law

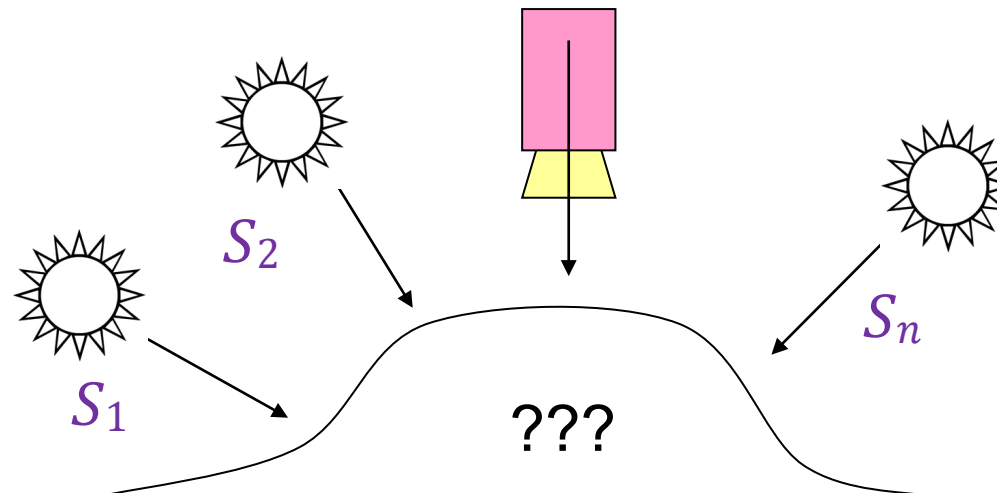
- What determines the observed brightness at a point on the surface?



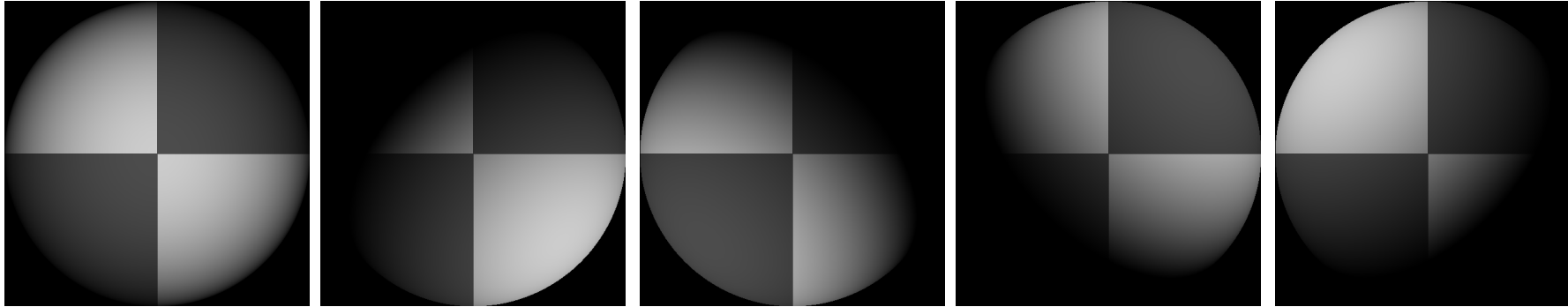
$$\begin{aligned} I &= \rho (S \cdot N) \\ &= \rho \|S\| \cos \theta \end{aligned}$$

Photometric Stereo

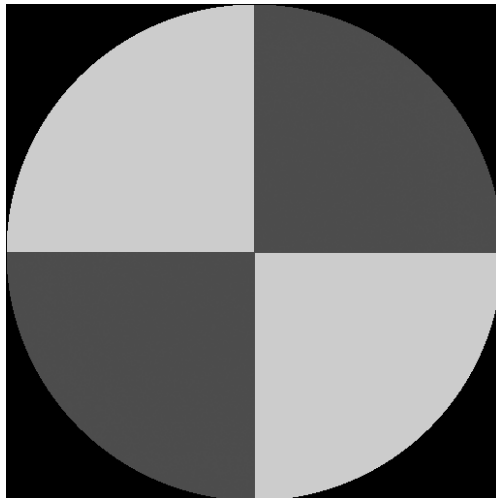
- Assume:
 - A Lambertian object
 - A *local shading model* (each point on a surface receives light only from sources visible at that point)
 - A set of *known* light source directions
 - A set of pictures of an object, obtained in exactly the same camera/object configuration but using different sources
 - Orthographic projection
- Goal: reconstruct object shape and albedo



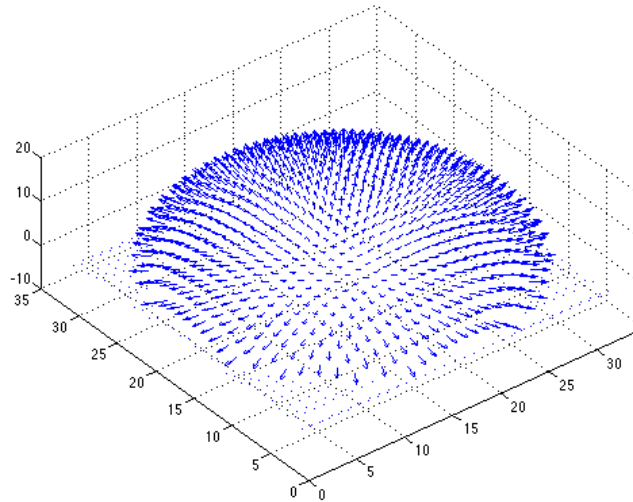
Synthetic Example



Recovered
albedo



Recovered normal
field



Recovered surface
model

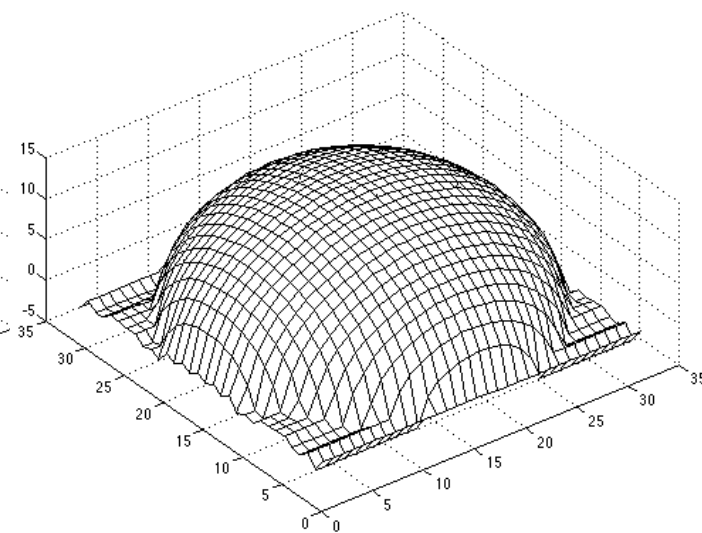


Image Model

- **Known:** source vectors S_j and pixel values $I_j(x, y)$
- **Unknown:** surface normal $N(x, y)$ and albedo $\rho(x, y)$

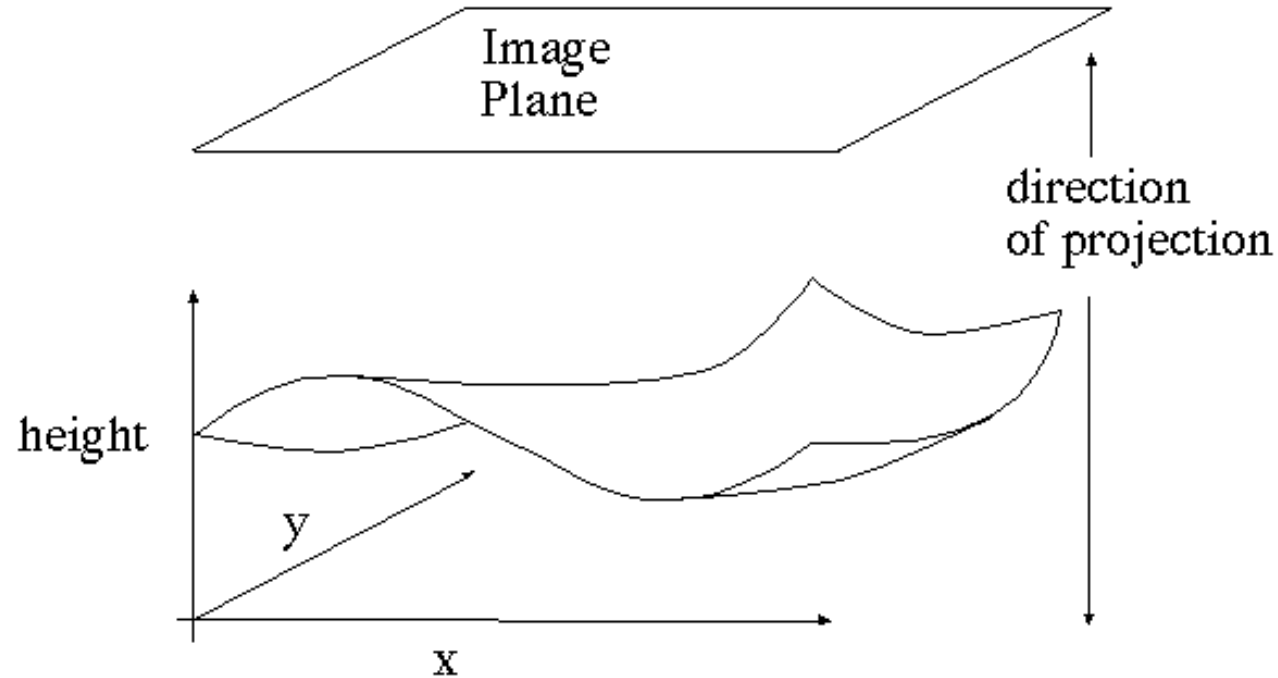


Image Model

- **Known:** source vectors S_j and pixel values $I_j(x, y)$
- **Unknown:** surface normal $N(x, y)$ and albedo $\rho(x, y)$
- Assume that the response function of the camera is a linear scaling by a factor of k
- Lambert's law:

$$\begin{aligned} I_j(x, y) &= k \rho(x, y) (N(x, y) \cdot S_j) \\ &= (\rho(x, y) N(x, y)) \cdot (k S_j) \\ &= g(x, y) \cdot V_j \end{aligned}$$

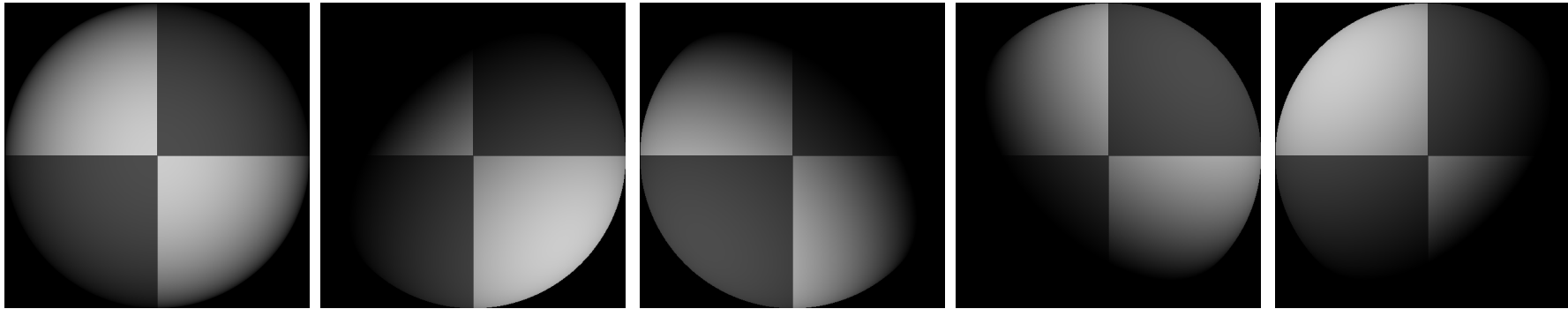
Least Squares Problem

- For each pixel, set up a linear system:

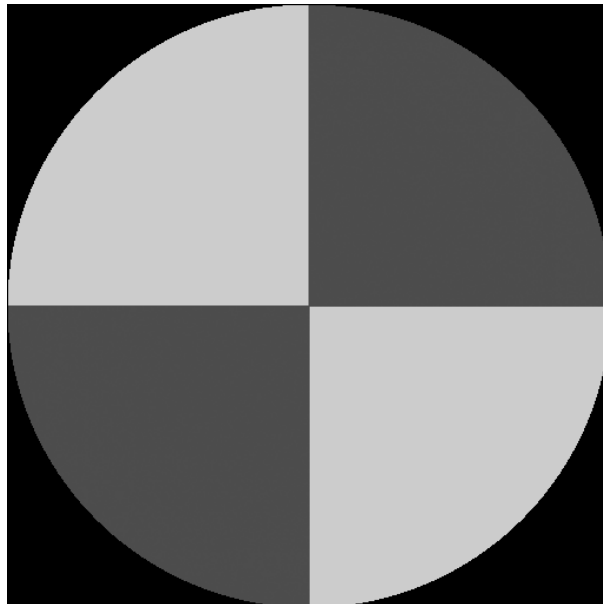
$$\begin{array}{ccc} \begin{bmatrix} V_1^T \\ V_2^T \\ \vdots \\ V_n^T \end{bmatrix} & \begin{array}{c} g(x, y) \\ | \\ 3 \times 1 \\ \text{unknown} \end{array} & = \begin{bmatrix} I_1(x, y) \\ I_2(x, y) \\ \vdots \\ I_n(x, y) \end{bmatrix} \\ n \times 3 \text{ known} & & n \times 1 \text{ known} \end{array}$$

- Obtain least-squares solution for $g(x, y)$, which we defined as $\rho(x, y)N(x, y)$
- Since $N(x, y)$ is the *unit* normal, $\rho(x, y)$ is given by the magnitude of $g(x, y)$
- Finally, $N(x, y) = \frac{1}{\rho(x, y)} g(x, y)$

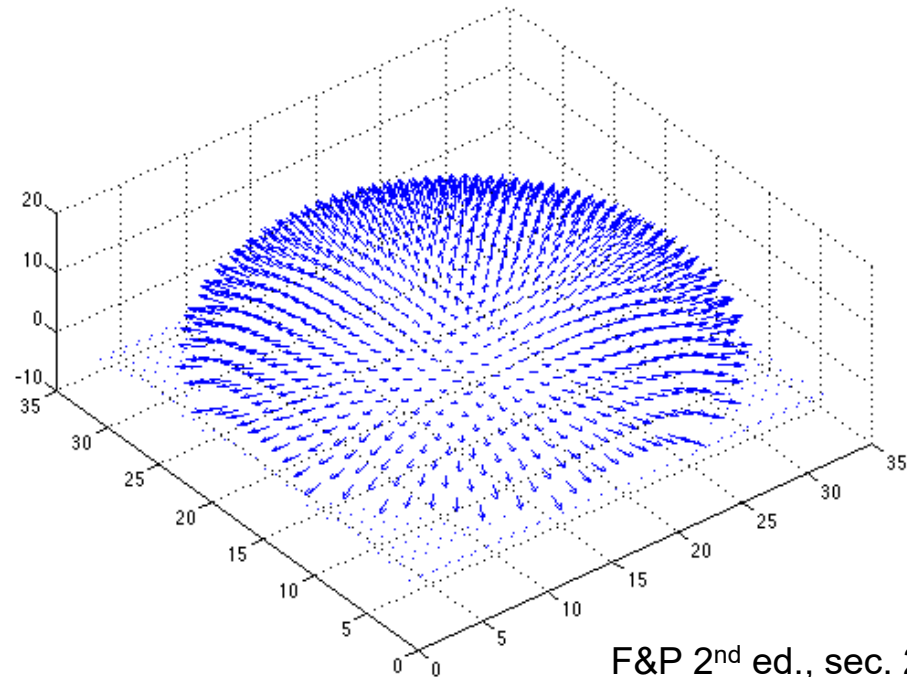
Synthetic Example



Recovered albedo



Recovered normal field



Recovering a Surface from Normals

- Recall: the surface is written as
- Write the estimated vector g as

$$(x, y, f(x, y))$$

$$g(x, y) = \begin{bmatrix} g_1(x, y) \\ g_2(x, y) \\ g_3(x, y) \end{bmatrix}$$

- This means the unit normal has the following form:
- Then we obtain values for the partial derivatives of the surface:

$$N(x, y) = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}} \begin{bmatrix} f_x \\ f_y \\ 1 \end{bmatrix}$$

$$f_x(x, y) = \frac{g_1(x, y)}{g_3(x, y)}$$

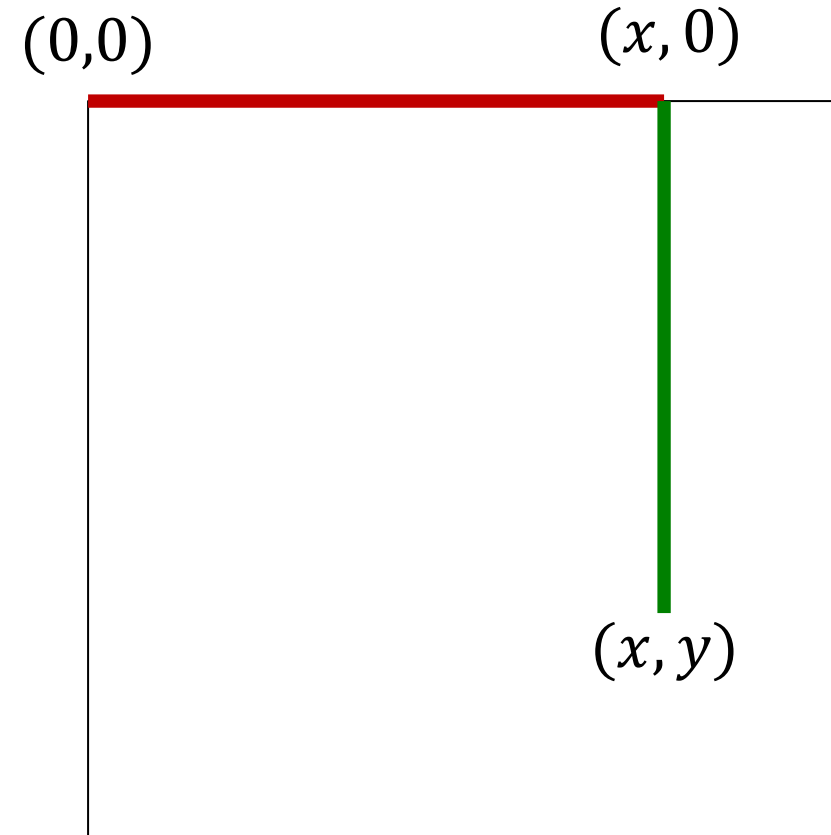
$$f_y(x, y) = \frac{g_2(x, y)}{g_3(x, y)}$$

Recovering a Surface from Normals

- We can now recover the surface height at any point by integration along some path, e.g.

$$f(x, y) = \int_0^x f_x(s, 0) ds + \int_0^y f_y(x, t) dt + C$$

- For robustness, it is better to take integrals over many different paths and average the results



Recovering a Surface from Normals

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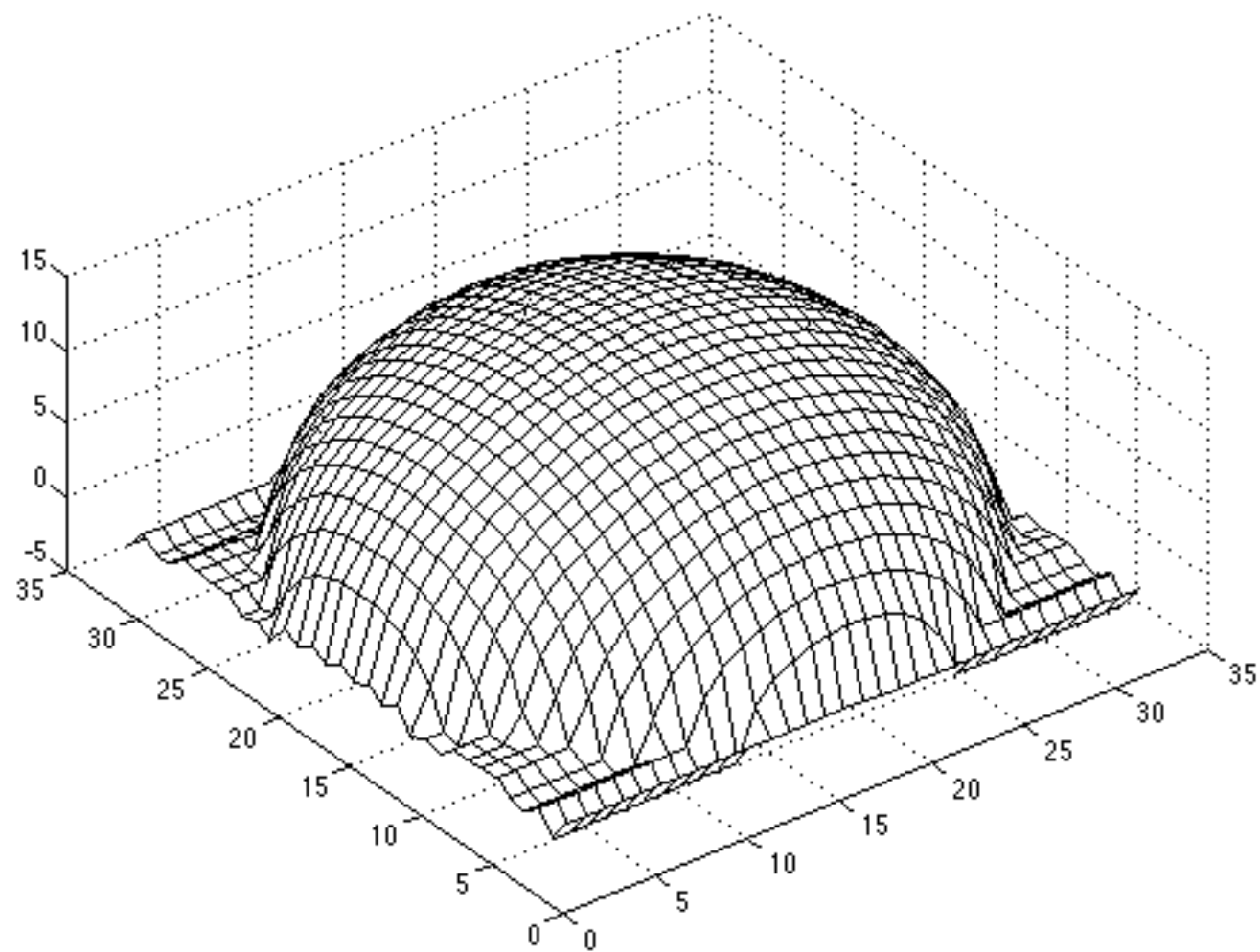
$$f(x, y) = \int_0^x f_x(s, 0) ds + \int_0^y f_y(x, t) dt + C$$

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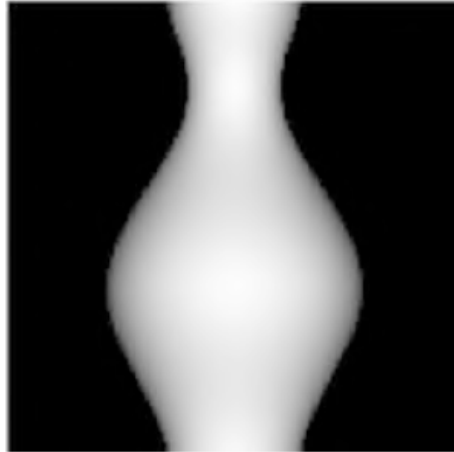
- Note: *integrability* must be satisfied: for the surface f to exist, the mixed second partial derivatives must be equal (or at least similar in practice):

$$\frac{\partial}{\partial y} \left(\frac{g_1(x, y)}{g_3(x, y)} \right) = \frac{\partial}{\partial x} \left(\frac{g_2(x, y)}{g_3(x, y)} \right)$$

Surface Recovered by Integration



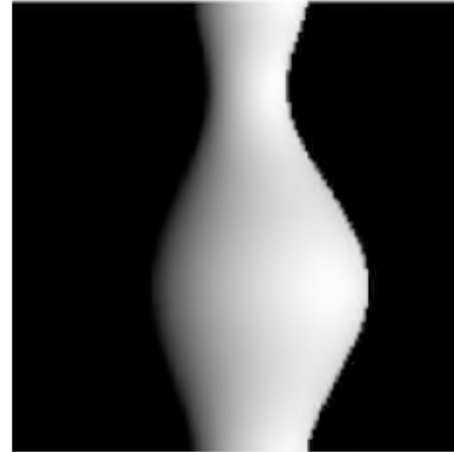
Shape from Shading



(a)



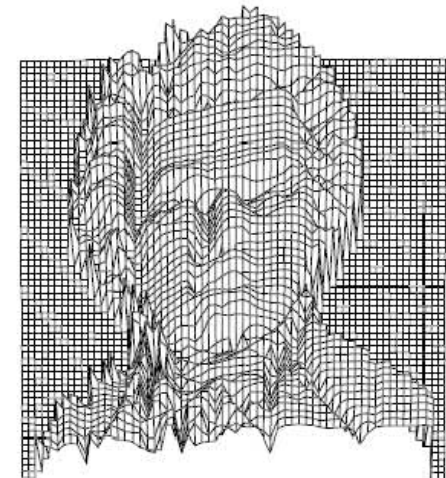
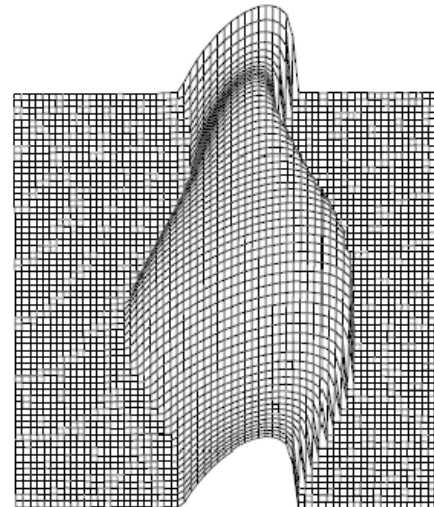
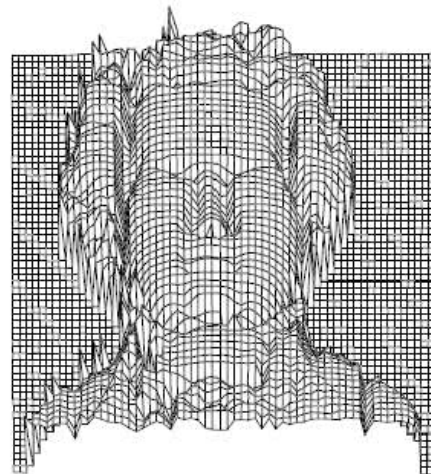
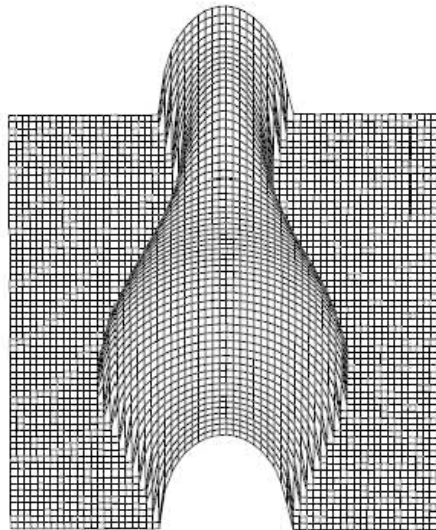
(b)



(c)



(d)

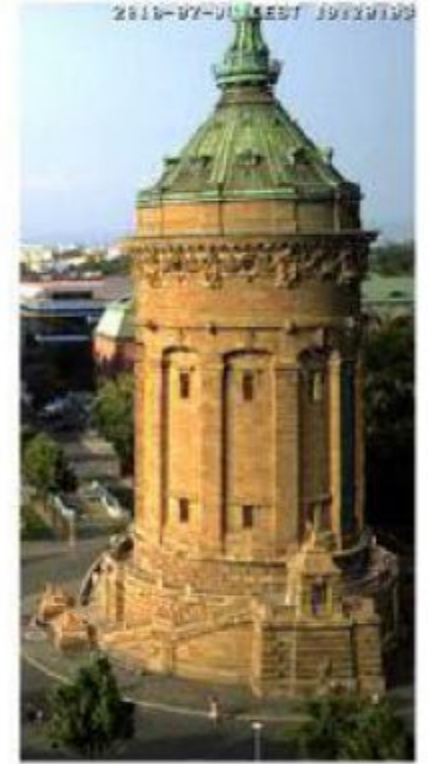
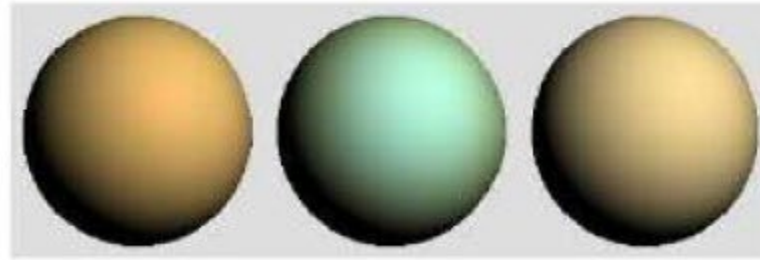
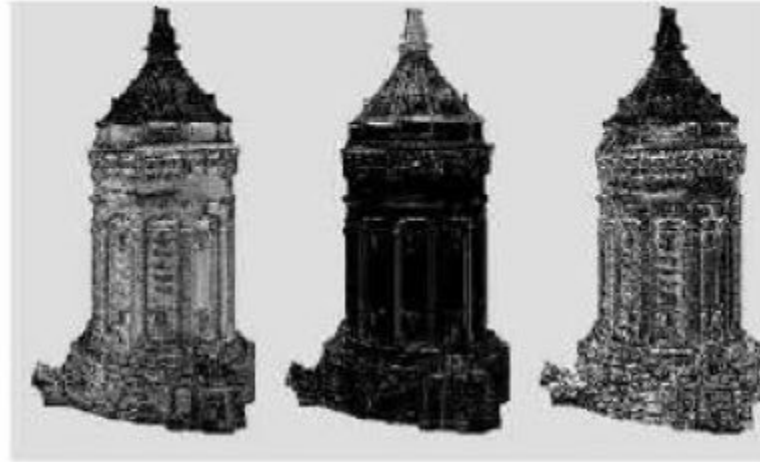


Multi-View Photometric Stereo



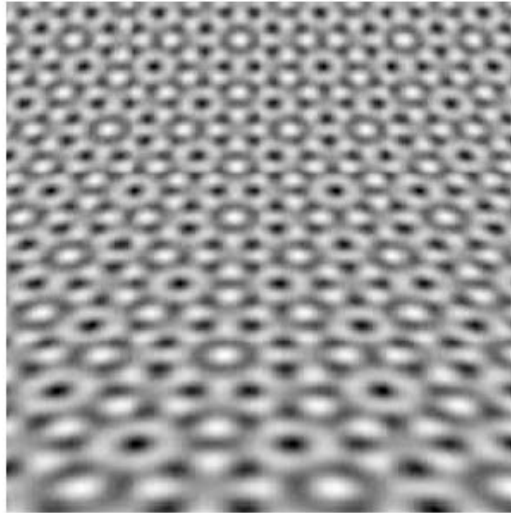
Initial multi-view stereo reconstruction refined

Webcam-based Outdoor Photometric Stereo

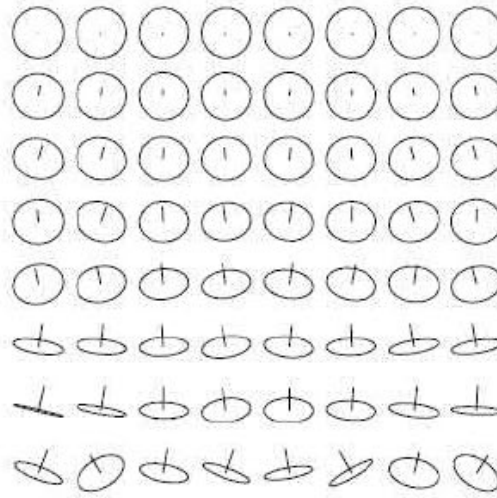


An input image, the recovered normal map, three basis BRDFs
Below their respective material maps, and a synthetic rendering
from a new sun position

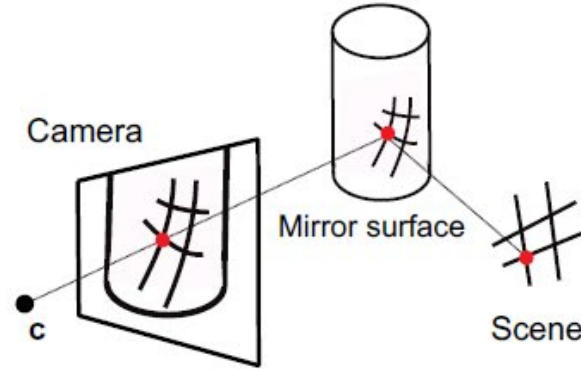
Shape from Texture



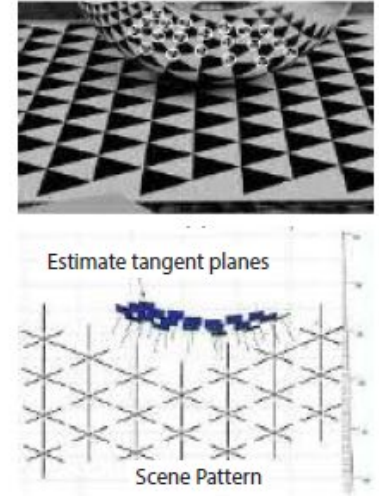
(a)



(b)



(c)



(d)

- (a) Regular texture wrapped onto a curved surface
- (b) The corresponding surface normal estimates
- (c) A regular pattern reflecting off a curved mirror
- (d) Curved lines, from which 3D point locations and normal can be inferred.

MP4 Preview

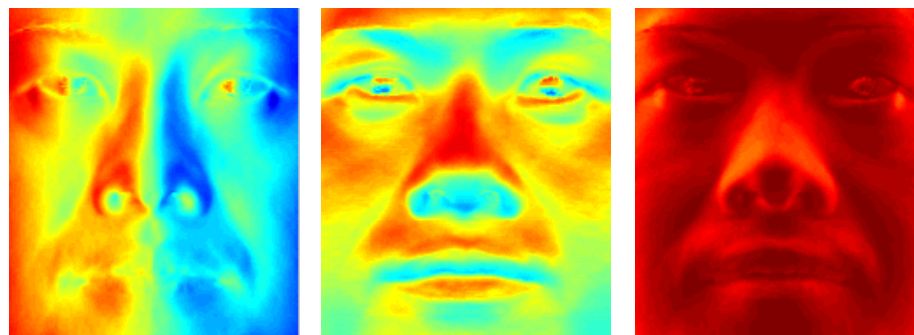
Input



Recovered
albedo



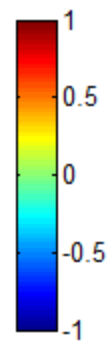
Recovered normal field



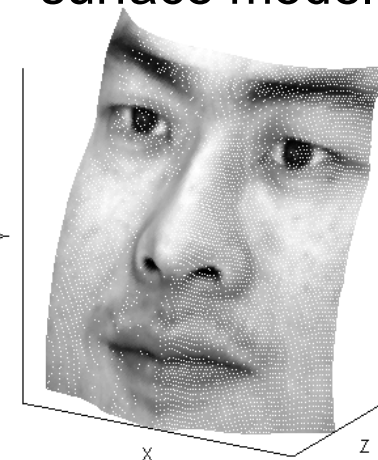
x

y

z



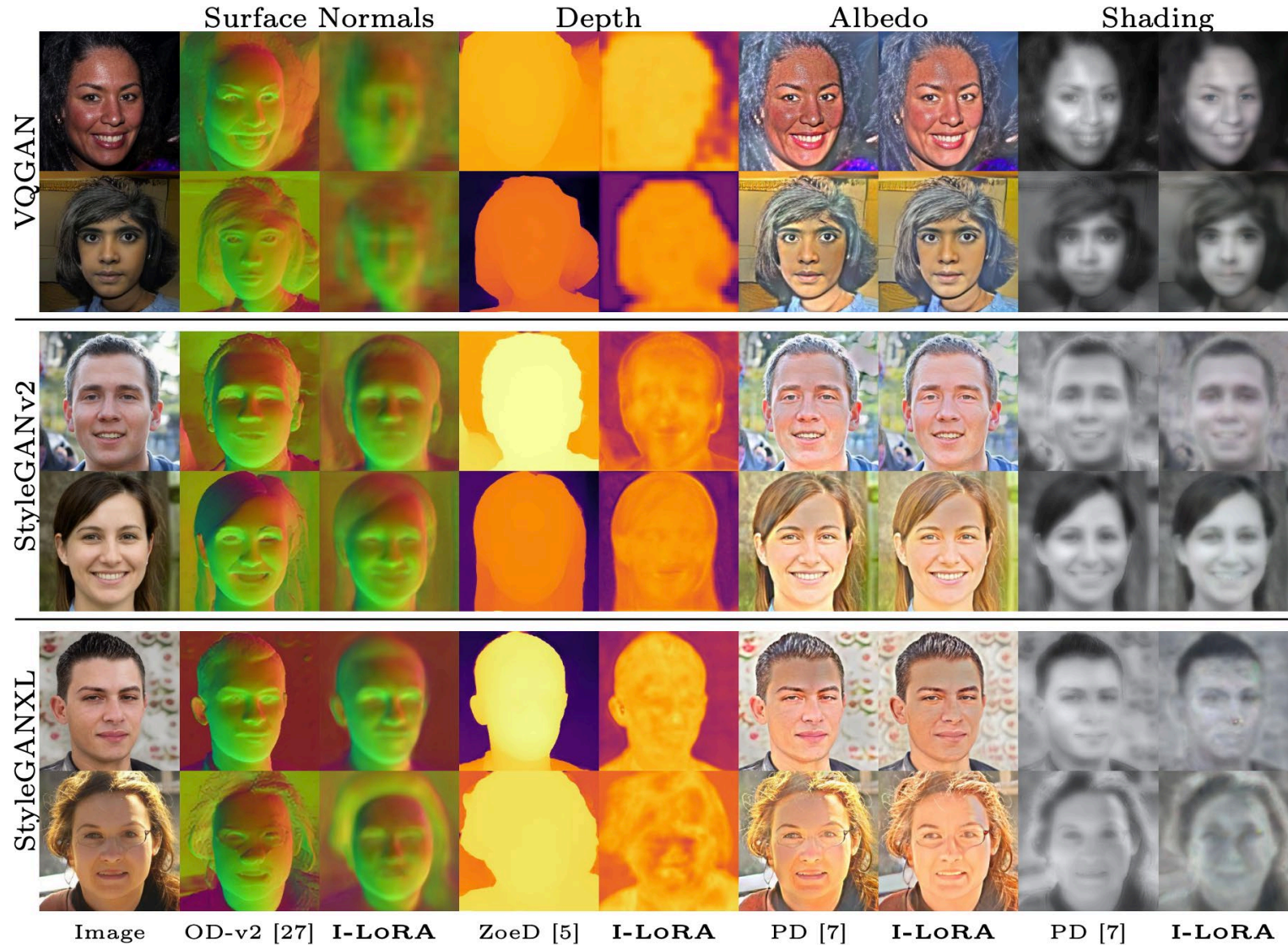
Recovered
surface model



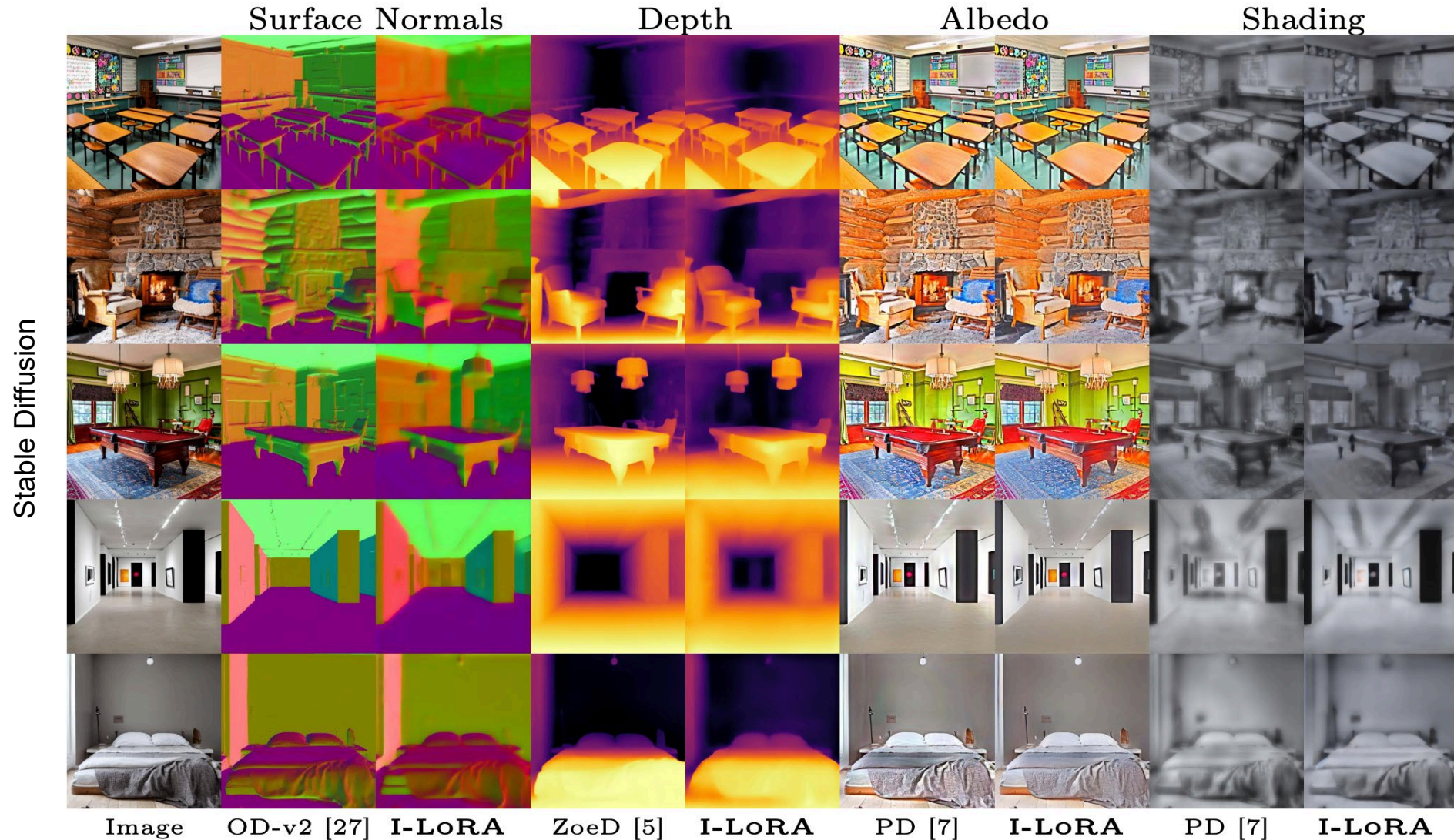
Limitations of Basic Shape from Shading

- Orthographic camera model
- Simplistic reflectance and lighting model
- No shadows
- No interreflections
- No missing data
- Integration is tricky

Shape from Shading Today



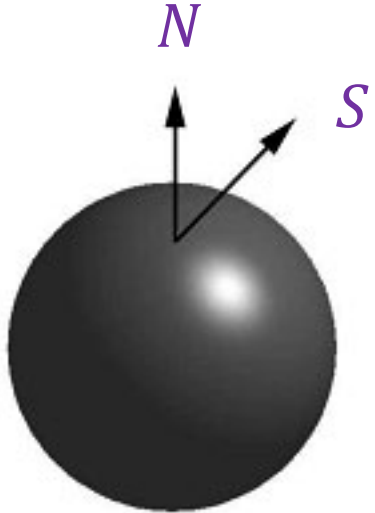
Shape from Shading Today



Estimating Direction of Light Sources

Finding the Direction of the Light Source

$$I(x, y) = N(x, y) \cdot S(x, y)$$

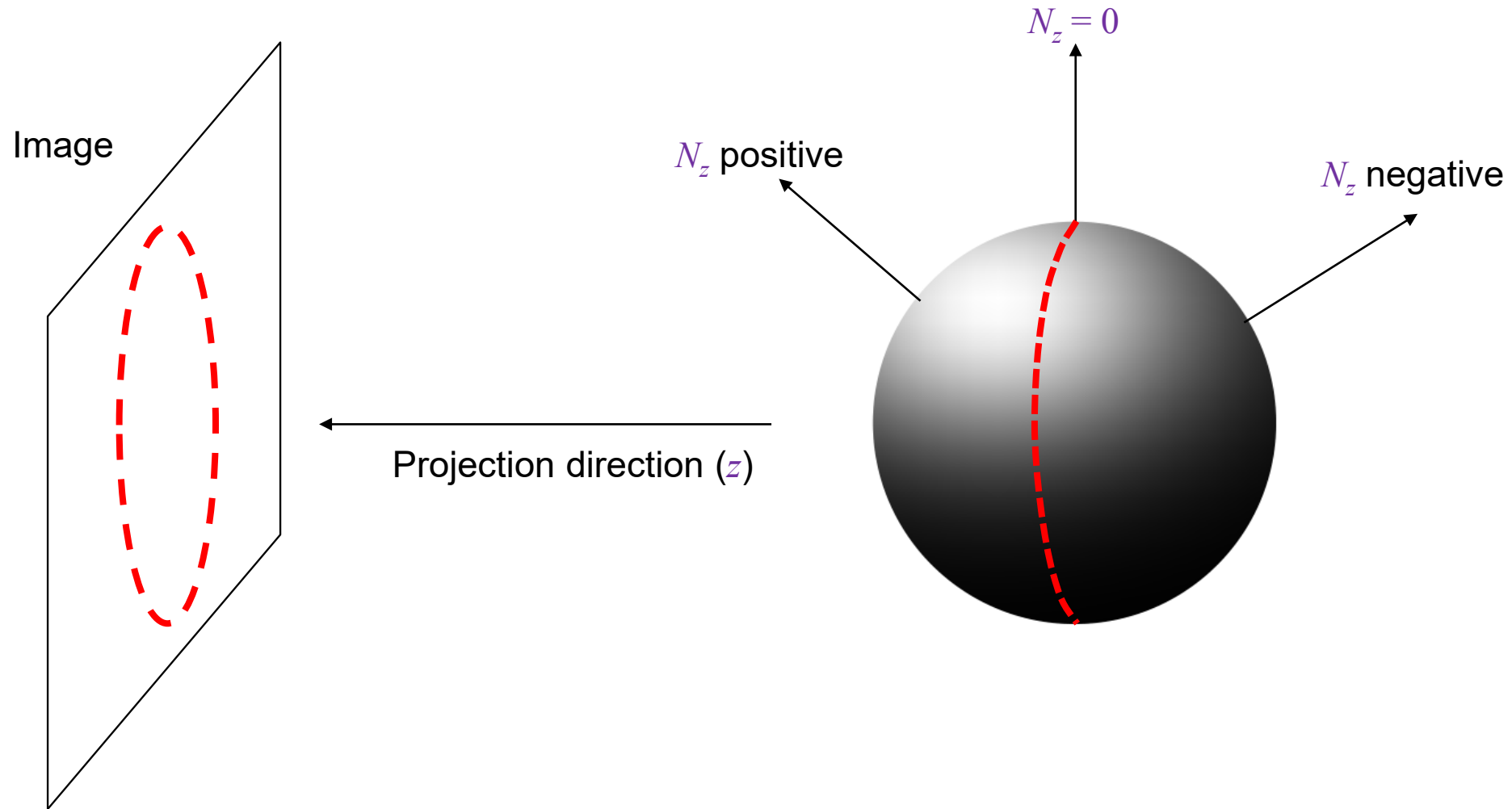


- Full 3D case:

$$\begin{bmatrix} N_x(x_1, y_1) & N_y(x_1, y_1) & N_z(x_1, y_1) \\ N_x(x_2, y_2) & N_y(x_2, y_2) & N_z(x_2, y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n, y_n) & N_y(x_n, y_n) & N_z(x_n, y_n) \end{bmatrix} \begin{bmatrix} S_x \\ S_y \\ S_z \end{bmatrix} = \begin{bmatrix} I(x_1, y_1) \\ I(x_2, y_2) \\ \vdots \\ I(x_n, y_n) \end{bmatrix}$$

Finding the Direction of the Light Source

Consider points on the *occluding contour*.



Finding the Direction of the Light Source

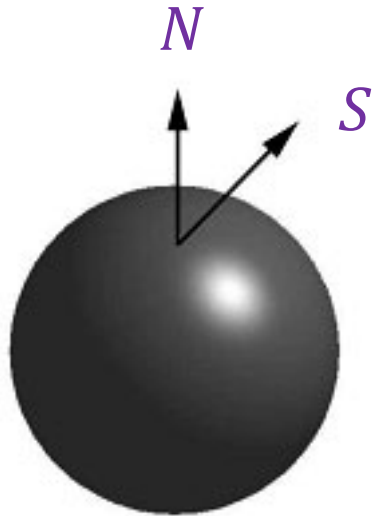
$$I(x, y) = N(x, y) \cdot S(x, y)$$

- Full 3D case:

$$\begin{bmatrix} N_x(x_1, y_1) & N_y(x_1, y_1) & N_z(x_1, y_1) \\ N_x(x_2, y_2) & N_y(x_2, y_2) & N_z(x_2, y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n, y_n) & N_y(x_n, y_n) & N_z(x_n, y_n) \end{bmatrix} \begin{bmatrix} S_x \\ S_y \\ S_z \end{bmatrix} = \begin{bmatrix} I(x_1, y_1) \\ I(x_2, y_2) \\ \vdots \\ I(x_n, y_n) \end{bmatrix}$$

- For points on the occluding contour ($N_z = 0$):

$$\begin{bmatrix} N_x(x_1, y_1) & N_y(x_1, y_1) \\ N_x(x_2, y_2) & N_y(x_2, y_2) \\ \vdots & \vdots \\ N_x(x_n, y_n) & N_y(x_n, y_n) \end{bmatrix} \begin{bmatrix} S_x \\ S_y \end{bmatrix} = \begin{bmatrix} I(x_1, y_1) \\ I(x_2, y_2) \\ \vdots \\ I(x_n, y_n) \end{bmatrix}$$



Finding the Direction of the Light Source



P. Nillius and J.-O. Eklundh. [Automatic estimation of the projected light source direction](#). CVPR 2001

Application: Detecting Composite Photos

Fake photo



Real photo



M. K. Johnson and H. Farid. [Exposing Digital Forgeries by Detecting Inconsistencies in Lighting](#).
ACM Multimedia and Security Workshop, 2005

DeepFake Detection Today



Generated Image

Shadow Errors

Detected Shadow Errors

Vanishing Point Errors

Detected Perspective Errors

A. Sarkar et al. [Shadows Don't Lie and Lines Can't Bend! Generative Models don't know Projective Geometry ... for now.](#)

CVPR 2024

Thank You